

A reusable solvent-response framework for molecular hydration prediction

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Code and data: <https://github.com/Scientific-Computing-Lab/SolvAI>

Abstract

Solvent response governs hydration free energy, yet high-fidelity calculations resolve it afresh for every new molecule. We ask whether information already encoded in disparate solvent datasets can instead be learned once and reused. SolvAI trains molecular surrogates on six calculated and empirical sources, predicts 15 complementary response descriptors from a SMILES string, and combines them with structural features in a separate predictor fitted to experimental hydration free energies. Inference requires no new solvent simulation. In a matched ExtraTrees comparison, the descriptors reduce mean absolute error from 0.303 to 0.202 kcal mol⁻¹ on 85 ARROW solutes and from 1.532 to 1.153 kcal mol⁻¹ on an endpoint-disjoint external cohort of 220 molecules. Shuffling descriptors among molecules removes the gain; it persists on the 97-molecule subset also absent from supervised response sources. The same descriptors often help other endpoint families, although the best complete predictor and the most useful source mixture change with the evaluation chemistry. Existing solvent calculations can therefore provide reusable supervision for molecular prediction, while source choice and transfer beyond small neutral solutes remain important tests.

Introduction

Hydration free energy is the reversible work of transferring a molecule from the gas phase into water. It connects molecular structure to partitioning, binding and reactivity, and is a stringent test of how a model balances electrostatics, polarization, dispersion, repulsion and solvent reorganization. Experimental values are available for many small neutral compounds, notably through FreeSolv, but predicting them with comparable precision remains difficult.^{1;2}

Physical simulation resolves solvent response directly. Alchemical methods vary the solute-water interaction along a coupling path and combine the resulting ensemble averages; path-integral molecular dynamics (PIMD) can also represent nuclear delocalization.^{3;4} The cost is that each new molecule calls for another sampling calculation. On 85 neutral solutes, the ARROW study made the accuracy at stake concrete: classical ARROW has a reconstructed mean absolute error (MAE) of 0.785 kcal mol⁻¹, falling to 0.205 kcal mol⁻¹ with eight-bead PIMD (PIMD8).⁵ Learned-potential extensions reproduce parts of that physical description, but still use trajectories at inference.^{6;7}

Prediction from molecular structure offers a different economy: train on earlier examples, then answer a new query without repeating their calculations. Graph and three-dimensional predictors,

Δ -learning and multi-fidelity methods use this principle in different ways.⁸⁻¹¹ Physically inspired descriptors can bring electrostatics, accessible surface or conformational response into such models.^{12;13} Yet approaches that calculate a continuum or molecular solvent response for each query retain a per-molecule physical step.¹⁴⁻¹⁷ Can that solvent information itself become a reusable molecular representation?

Predicting a physical descriptor before fitting a downstream property is an established route: quantum-chemical descriptors have been learned for reactivity, and a two-stage hydration model has predicted solute descriptors instead of calculating them for every query.^{18;19} The usefulness of such a descriptor depends on its relevance, how accurately it can be predicted and how the downstream model uses it.^{20;21} Here the question is whether several complementary kinds of solvent information, learned from different sources, remain useful across hydration predictors and unfamiliar molecules.

SolvAI uses the existing datasets as six groups of teachers (Fig. 1). They supply 15 targets spanning calculated transfer free energies, empirical solute properties and summaries of how water changes conformer energies. Molecular surrogates learn to predict these quantities from structure and are frozen before a separate endpoint learns experimental hydration free energy. We call each predicted quantity a *response descriptor*; the name does not imply a derivative with respect to an applied field. Because the two learning stages are separate, we can change the hydration predictor while holding the descriptors fixed, or change their surrogates while holding the source targets fixed.

We first ask whether the predicted responses improve a matched hydration predictor and whether the gain survives controls that break molecule alignment or separate training from test chemistry. We then hold the response representation fixed and vary the endpoint, distinguishing the value of adding responses to a given learner from the accuracy of the complete learner. Finally, we vary the information used to build the representation and test its behavior on larger molecules. The later architecture comparisons are retrospective; their selection protocol is detailed in Methods.

SolvAI: learning solvent response before hydration free energy

Two learning tasks with different supervision

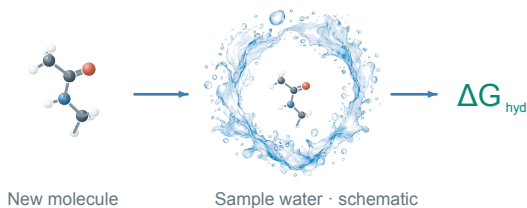
For a neutral molecule represented by a SMILES string x , the target is its experimental gas-to-water transfer free energy $y = \Delta G_{\text{hyd}}$ near 298 K. We first train separate molecular models on external source datasets. Each model predicts one or more source quantities, such as a continuum-solvent free energy or a hydrogen-bonding parameter. These models are *surrogates*: after training, they approximate the source quantities without repeating the source calculation or measurement. Their weights are frozen before learning the experimental hydration predictor.

Writing the frozen surrogate predictions as $\hat{r}_j(x)$, our model is

$$\hat{\mathbf{r}}(x) = (\hat{r}_1(x), \dots, \hat{r}_{15}(x)), \quad \hat{y}(x) = h([\mathbf{s}(x), \hat{\mathbf{r}}(x)]), \quad (1)$$

where $\mathbf{s}(x)$ denotes structural features; the standard tabular input contains a 2,048-bit Morgan fingerprint and 217 RDKit two-dimensional descriptors. The endpoint h learns how the predicted source quantities relate to experiment; it does not sum them as if they were components of one physical free energy. The primary comparisons keep the same 15 frozen predictions while varying h and, in some models, the structural representation (Table 2). All 15 quantities are predicted at deployment, including those whose original source targets were experimental.

a The cost of a new solvent calculation



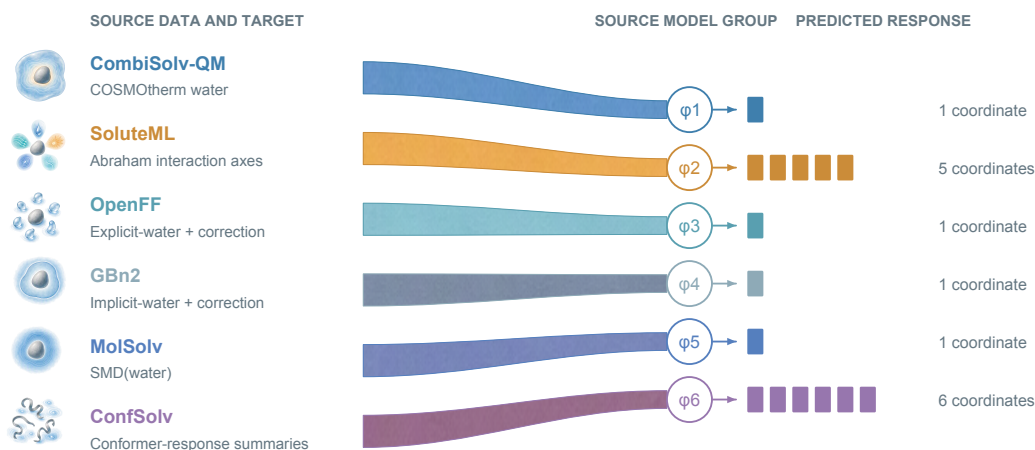
A new physical calculation is costly

when it must be repeated for every query.

SolvAI learns from existing solvent information.

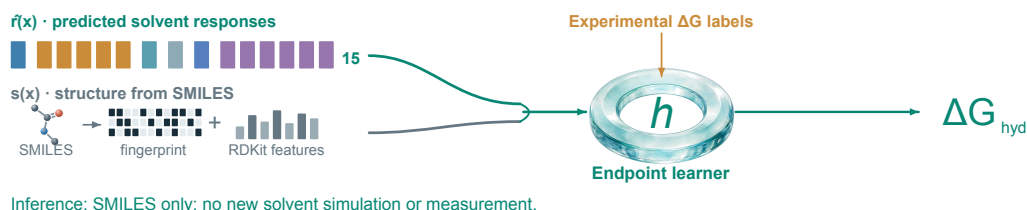
b Six source datasets teach reusable response maps

Different targets; one common molecular input



Freeze six model groups → 15 predicted response coordinates

c Learn hydration from experimental labels



d Evaluation separates endpoint and source exposure

ARROW-85



85 absent from all six supervised source sets
Endpoint: out of fold

External-220



220 endpoint-disjoint
■ 123 source-exposed
■ 97 source-disjoint

Figure 1: Existing solvent information becomes reusable molecular supervision. **a**, Explicit-solvent hydration calculations sample a water ensemble for each molecule to estimate ΔG_{hyd} . The molecular drawing and solvent halo are schematic, not simulation output. **b**, Six source datasets teach separate structure-to-response model groups, frozen before endpoint fitting. Their 1 + 5 + 1 + 1 + 1 + 6 predicted coordinates total 15; a group can contain target-specific fits. OpenFF and GBn2 combine a physical-energy prediction with a learned experimental residual. Icons and colored ribbons distinguish sources; they do not depict measured response fields or magnitudes. **c**, A separate hydration endpoint learns from predicted responses and SMILES-derived structural features using experimental hydration labels. The molecule, fingerprint and RDKit-feature glyphs show input types, not actual feature values. Inference requires only SMILES, with no new solvent calculation. **d**, Each tile represents one evaluation molecule: 85 source-disjoint ARROW molecules scored out of fold; 220 endpoint-disjoint external molecules, including 123 source-exposed and 97 source-disjoint. Strict-97 is nested in External-220. Identity checks include standardized equivalents. The diagram is architecture-neutral; source learners and endpoints are compared in Supplementary Table 17 and Table 2.

What the response descriptors represent

The sources offer different views of solvation rather than successive rungs on one fidelity scale (Table 1; Fig. 1b). Continuum-solvent calculations estimate transfer energetics; Abraham parameters describe empirical interaction tendencies; corrected explicit- and implicit-solvent estimates combine a physical prediction with an experimental residual. ConfSolv adds information about how water redistributes conformer energies and solvation responses. Together they give the endpoint several kinds of evidence about a molecule, rather than asking one source to stand for the whole solvent problem. The source methods and their physical definitions are documented in the original literature and Supplementary Methods S3 and Table 1.

Table 1: The 15 response descriptors and the information they provide. Every listed quantity is predicted from molecular structure at inference. CombiSolv-QM supplies COSMOtherm targets, SoluteML supplies Abraham parameters, and MolSolv supplies SMD(water) targets. The last column describes the released implementation, not an architectural requirement of the framework. Complete definitions, units and source counts are in Supplementary Table 1.

Source	Number	Predicted quantity	Released surrogate
CombiSolv-QM	1	COSMOtherm water transfer free energy	D-MPNN
SoluteML	5	Abraham E, S, A, B, L interaction parameters	ExtraTrees
OpenFF	1	Explicit-solvent estimate plus experimental residual	ExtraTrees
GBn2	1	Implicit-solvent estimate plus experimental residual	ExtraTrees
MolSolv	1	SMD continuum-water transfer free energy	D-MPNN
ConfSolv	6	Conformer corrections and water-response distributions	LightGBM

The source archives contain many calculations per molecule. MolSolv’s 1,729,545 records are SMD(water) calculations; ConfSolv’s 5,392,567 are water-conformer records. After quality, target and identity filtering, the respective surrogate tables retain 350,359 and 17,829 molecular connectivities. Archive records and training rows are different units of counting (Supplementary Table 1 and Fig. 2).

Hydration free energy is an equilibrium property of a molecule, not a separate experimental label for each constrained geometry. ConfSolv’s six coordinates summarize conformational response without reconstructing an ensemble at inference. Supplementary Fig. 7 illustrates this distinction, which also matters when comparing the response features with Uni-Mol’s generated three-dimensional conformer.

Results

Separating source supervision from hydration evaluation

Two experimental settings ask different questions. ARROW-85 provides experimental hydration values and a published high-fidelity simulation result for the same small molecules; its folds test the learned predictor out of sample. A separate cohort of 220 neutral solutes, drawn from original Henry-law measurements, tests prediction outside the endpoint training labels. All comparisons of SolvAI in this paper use experiment as the reference; the ARROW simulation is an accuracy context, not a model input.

The endpoint training pool contains 1,280 public hydration labels plus available ARROW training folds. The response-source tables exclude every exact or standardized equivalent of the 85 ARROW test solutes. All 220 external molecules are absent from endpoint labels, and a nested set of 97 is absent from the supervised response sources as well. We distinguish

these two forms of exposure throughout because a source calculation for a molecule is not its experimental hydration label. Exclusions and affected surrogate refits are documented in Methods, Supplementary Table 2 and Supplementary Data 4.

A matched test of reusable response information

We begin with a matched question: do the predicted responses add information when the hydration learner stays the same? An ExtraTrees ensemble of randomized decision trees³⁰ is trained with or without the 15 response columns, using identical labels, folds, structural features and training recipe. Each ARROW solute is predicted out of fold by a fit that did not receive its experimental label. Structural features alone yield an MAE of 0.303 kcal mol⁻¹; the first eight response descriptors reduce it to 0.212, SMD(water) to 0.207, and the six conformer summaries to 0.202 kcal mol⁻¹ (Fig. 2b). That final value is numerically close to the published ARROW/PIMD8 error of 0.205 on these solutes. PIMD8 is a separate physical simulation, while this endpoint learns from experimental hydration labels; the matched test here is the comparison with and without response descriptors.

The gain is distributed across molecules rather than driven by a single large correction: absolute error falls for 53 of 85 solutes, and the paired MAE change is -0.101 kcal mol⁻¹ (95% bootstrap interval, -0.215 to -0.020; Fig. 2c). The full set also improves more than the tested computed-energy, SMD-only or ConfSolv-only blocks in isolation (Fig. 2d). The empirical and residual-corrected block contributes on its own, suggesting that the sources supply partly complementary information. Molecule-level predictions and residuals are in Supplementary Fig. 1 and Data 2.

Shuffling tests whether the benefit depends on assigning the responses to the correct molecule (Fig. 2a). Across five independent permutations of the 15-column block, the fixed-partition mean is 0.306 kcal mol⁻¹, close to the 0.303 matched control and far from the aligned-response fit. The paired difference between aligned and mean-shuffled descriptors is -0.104 kcal mol⁻¹ (95% interval, -0.214 to -0.025). Alignment between a molecule and its predicted response is therefore necessary for the observed gain.

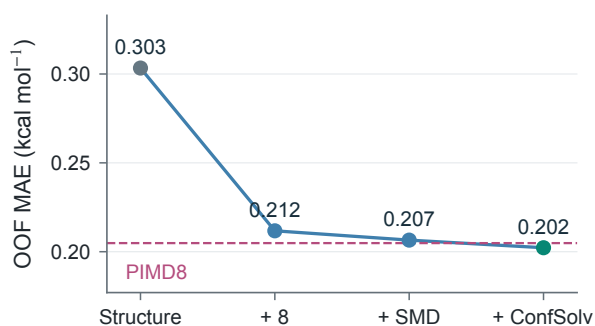
Changing the ARROW partition changes the absolute estimate but preserves the matched contrast. Across five complete five-fold partitions, the response-augmented endpoint has an average MAE of 0.207 ± 0.004 kcal mol⁻¹, versus 0.313 for structure alone, and improves in every partition (Fig. 2e; Supplementary Table 4). Reducing the preset weight of available ARROW training labels from 3 to 1 also preserves the contrast (Supplementary Table 9).

a Change only the response information

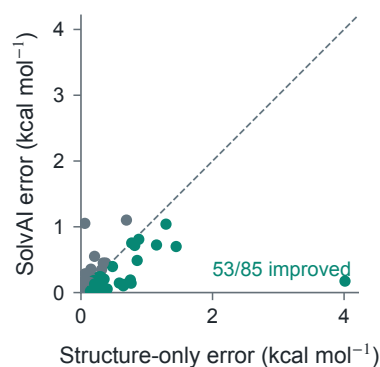
Same structures, labels, folds, weights, learner and seeds



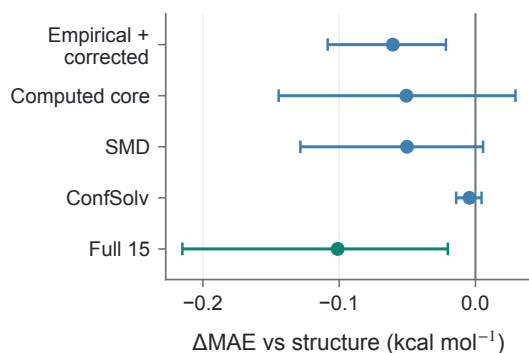
b Build the response set



c Compare individual molecules



d Which source blocks contribute?



e Repeat the full partition

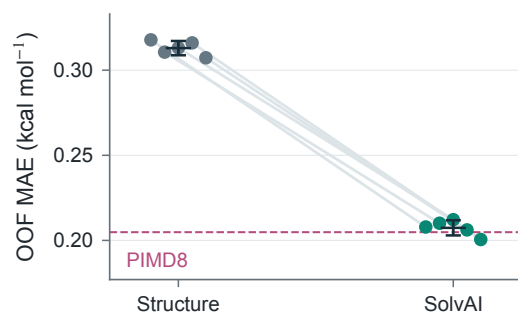


Figure 2: Molecule-aligned response descriptors improve a matched hydration endpoint. **a**, The matched control removes the responses; the shuffled control reassigns the complete response block among molecules. Labels, structural descriptors, architecture, weights, folds and seeds are fixed. The assignment sketches are illustrative; shuffled MAE is the mean of five permutations. **b**, Fixed-partition progression adds eight COSMOtherm, Abraham and corrected-solvent descriptors, then SMD(water), then six ConfSolv summaries. The dashed magenta line marks ARROW/PIMD8 on the same 85 solutes, as an accuracy reference rather than an input or training source. **c**, Paired absolute errors in kcal mol⁻¹ for every molecule. Points below the diagonal mark lower error with responses (teal); gray points do not improve. **d**, MAE change for predeclared response blocks relative to structure only; bars are 95% molecule-bootstrap intervals from 100,000 resamples. **e**, Paired results for five complete split repeats. Thin lines join methods within the same partition; black marks show mean ± sample s.d.

Does the response help beyond one endpoint?

The matched test raises a second question: does the information in the 15 frozen descriptors remain useful when the hydration learner changes? We tested nine fixed endpoint configurations, from trees and neural networks to molecular graphs and pretrained encodings (Table 2). For each one, a paired fit removes only the response columns while retaining its selected structural inputs and training recipe. Thus the comparison asks both how accurate the complete learners are and what the responses contribute within each learner. TabM uses a shared-parameter neural ensemble, and TabPFN predicts from a pretrained transformer conditioned on labelled training rows.^{31–33} The study is retrospective; configurations were selected using training-side validation, with the full protocol in Methods and Supplementary Methods S18.

Table 2: Endpoint mechanisms and their measured accuracy. All entries include the 15 response descriptors. MAE is in kcal mol^{−1}. Boldface marks the lowest displayed MAE in each column among these fixed full-response configurations; it does not imply a statistically resolved rank. The first three columns use the ARROW and external evaluation identities; Strict-97 is nested within External-220. Neural and TabPFN predictions average five seeds, tree controls three. The final column is a separate exploratory size-held-out refit on 1,160 training and 205 held-out molecules, using three seeds for every method. A dash means not evaluated, not zero error. Supplementary Table 13 includes matched no-response controls and common-three-seed results.

Endpoint	Mechanism and observed pattern	ARROW	External	Strict	Size
ExtraTrees	Leaf averages; bounded output	0.202	1.153	1.536	1.772
Residual MLP	Residual links; external response gain	0.340	1.428	2.016	1.813
Dual branch	Separate inputs; improved size transfer	0.288	1.234	1.616	1.560
Ridge + neural residual	Linear trend + correction; low Strict error	0.336	1.172	1.429	1.391
Message passing	Size-aware graph; external response gain	0.288	1.168	1.505	1.729
MoLFormer + neural head	Frozen sequence encoding; low external error	0.273	1.111	1.450	1.609
TabM	Shared neural ensemble; low size error	0.291	1.132	1.454	1.389
TabPFN-3.5	Label-conditioned transformer; low size error	0.199	1.134	1.493	1.318
MoLFormer + ExtraTrees	Frozen encoding + trees; input contrast	0.234	1.174	1.564	–

The lowest full-model point estimate changes with the cohort: TabPFN reaches 0.199 kcal mol^{−1} on ARROW-85, a MoLFormer neural head 1.111 on External-220, and ridge plus a neural residual 1.429 on Strict-97 (Table 2). These are descriptive rankings of complete predictors. They cannot tell us whether the same predictor needed the response descriptors.

The paired fits answer that question (Fig. 3). On External-220 and its nested Strict-97 subset, seven of nine configurations have below-zero descriptive intervals for the response-minus-no-response MAE difference. The neural MoLFormer endpoint, for example, changes from 1.332 to 1.111 externally; ExtraTrees, message passing and several other learners also benefit. On ARROW-85, the intervals are below zero for ExtraTrees and the two MoLFormer endpoints. The contrast has exceptions that make the distinction useful: TabPFN has the lowest ARROW point estimate without a resolved response increment, while the lowest full-model

159 Strict-97 configuration, ridge plus a neural residual, has lower error when its responses are
 160 removed. The representation can therefore help across architectures without defining a single
 161 best complete model. Complete paired results, mechanism diagrams and selection variants are
 162 in Supplementary Tables 13–15, Figs. 8–10 and Data 6.

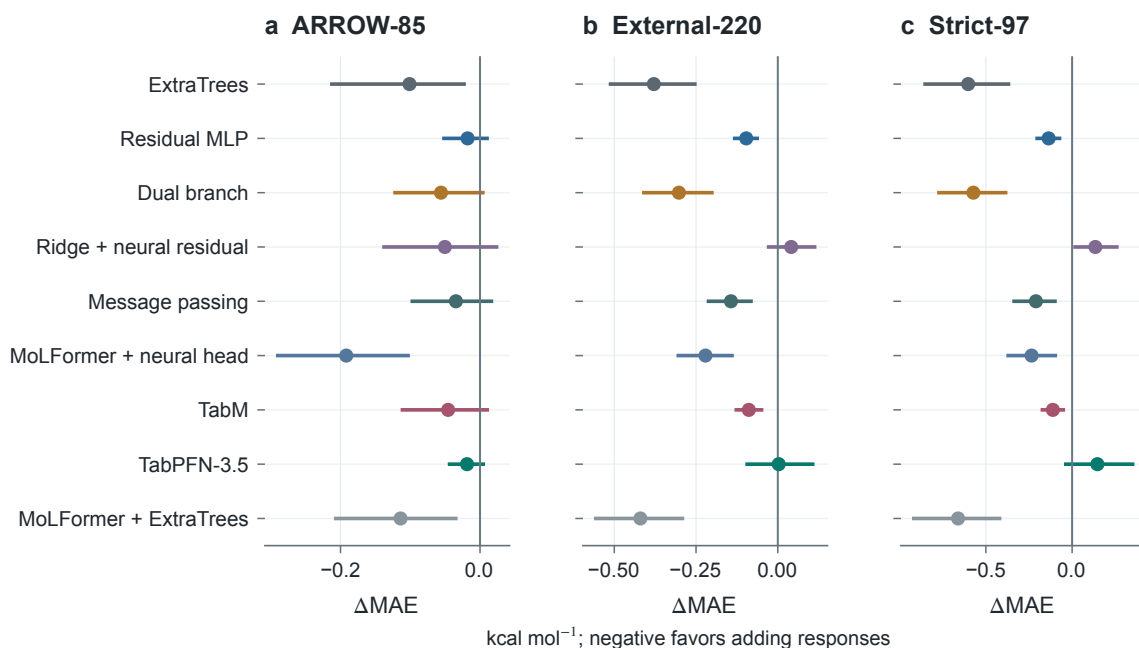


Figure 3: The response contribution varies among endpoint families. a–c, Paired MAE changes from adding the same 15 response descriptors to each selected endpoint recipe on ARROW-85, External-220 and Strict-97. Negative values favor response augmentation. Bars are descriptive 95% percentile intervals from 100,000 paired molecule resamples, conditional on the fitted models and without multiplicity correction. They are not uncertainty intervals for individual molecular predictions. Strict-97 is nested within External-220. Horizontal scales differ between cohorts. The no-response controls inherit the full model’s recipe. Cross-architecture accuracy and seed counts are given in Table 2; family-specific configuration choices are retained in Supplementary Data 6.

163 The responses might help only because the structural input to ExtraTrees is limited. A focused
 164 comparison holds that endpoint fixed while adding the 15 descriptors to frozen MoLFormer
 165 sequence encodings or Uni-Mol three-dimensional encodings (Fig. 4a). Direct graph predictors
 166 are shown as separate references; they are not the paired ablation for either frozen encoding.

167 The added responses lower ARROW-85 MAE from 0.348 to 0.234 kcal mol^{-1} with MoLFormer
 168 features and from 0.334 to 0.220 with Uni-Mol features. Both paired advantages also persist on
 169 External-220 and Strict-97, with intervals below zero (Fig. 4b,c). Thus the predicted solvent
 170 quantities provide useful information even beside pretrained molecular representations. The
 171 direct graph comparisons and their training protocols are reported in Methods and Supplementary
 172 Tables 11–12; the ten-descriptor source control shown in Fig. 4 is examined with the full source
 173 set below.

a Test alternative molecular representations

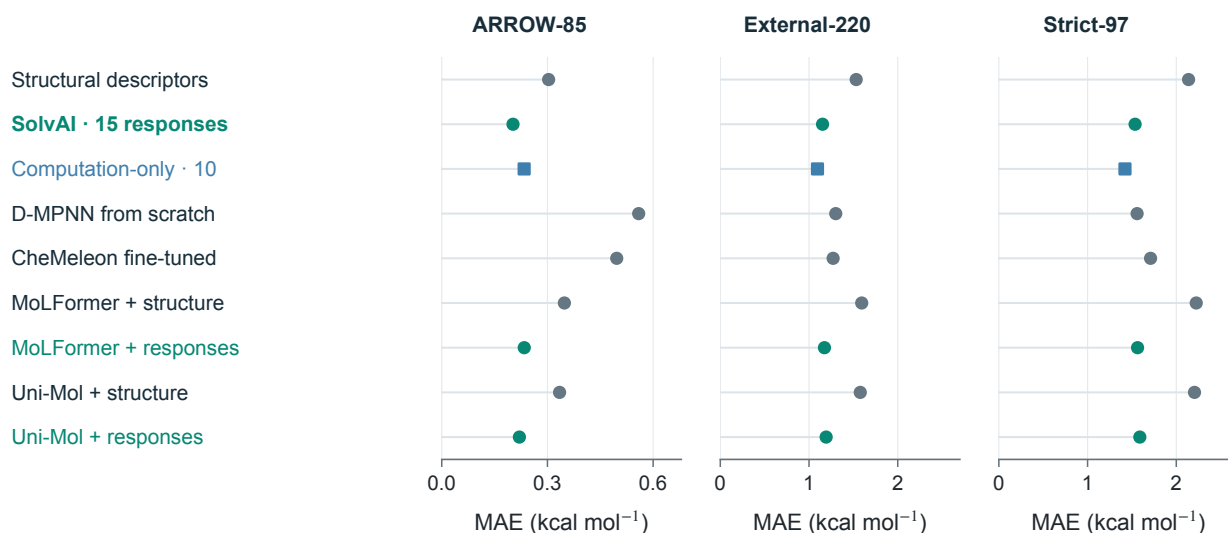
Direct graph learning



Frozen encoder + matched ExtraTrees



b Compare every model on the same held-out molecules



c Add responses to each fixed representation

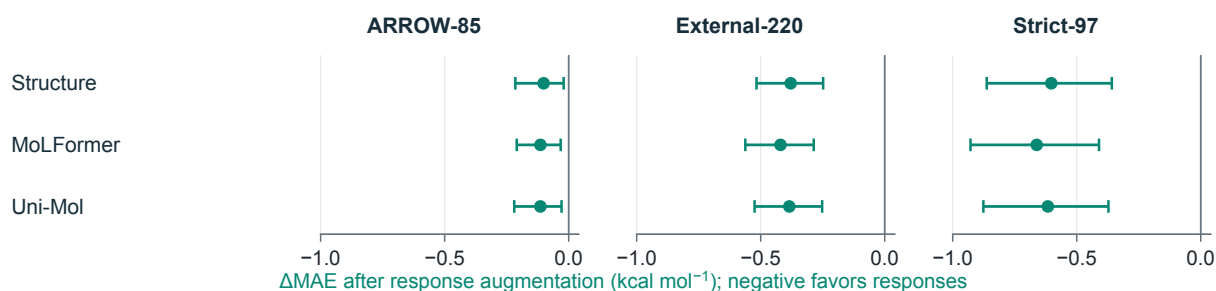


Figure 4: Response descriptors complement pretrained molecular representations. **a**, Direct graph models fit the entire network. Frozen MoLFormer and Uni-Mol representations instead augment the same structural features and ExtraTrees endpoint; paired fits add or remove only the 15 responses. **b**, All nine model conditions on ARROW-85, External-220 and Strict-97. Values are MAE in kcal mol⁻¹; note the different ARROW and external horizontal scales. ARROW predictions are out of fold. External predictions use all 1,365 endpoint-training labels; Strict-97 is a subset of External-220, not a separate fit. The ten-response control removes the Abraham parameters and experimental residual corrections (Fig. 6a). **c**, Paired changes after response augmentation, relative to each unaugmented representation. Negative values favor adding responses; bars are 95% intervals from 100,000 paired molecule resamples. Full numerical results and intervals are retained in Supplementary Tables 11–12.

Transfer beyond nearby labels

Random molecular folds can place close chemical neighbors on opposite sides of a train–test split. We therefore repeated the ExtraTrees comparison after excluding endpoint-training molecules in the test family’s scaffold, functional family or molecular cluster, and after removing neighbors above fixed Morgan-similarity thresholds. Each rule applies to the entire endpoint-supervised pool; the response surrogates remain fixed.

Adding responses lowers MAE under every predeclared separation rule (Fig. 5a,b; Supplementary Table 5). At a similarity exclusion threshold of 0.70, the matched error falls from 0.343 to 0.211 kcal mol^{−1}. Scaffold and functional-family splits are harder in absolute terms, yet their paired intervals also favor the response features. The gain therefore persists without nearby endpoint labels; it does not make these difficult splits equally accurate. Full regime values and molecule-level family errors are in Supplementary Tables 5–6 and Fig. 5.

Removing all ARROW experimental labels from endpoint training asks whether the representation needs fold-local adaptation to those 85 solutes. Trained only on the 1,280 public labels, the structure-only endpoint has an MAE of 0.385 kcal mol^{−1}, versus 0.257 with responses (paired interval for the change, −0.246 to −0.037; Fig. 5c). This is an ARROW-label removal test; the separate external cohort tests a more distinct set of molecules.

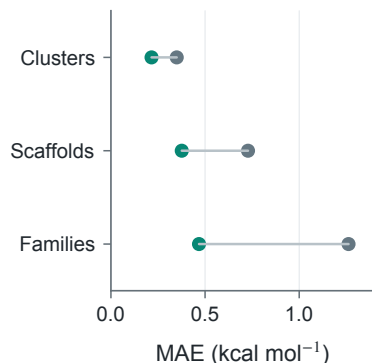
The external cohort supplies the stronger transfer test. It was frozen from original Henry-law measurements for a separate solvation-model study before that study’s predictions were examined.³⁴ On its 220 endpoint-disjoint neutral solutes, responses lower matched MAE from 1.532 to 1.153 kcal mol^{−1} (paired change −0.379; 95% interval, −0.517 to −0.248; Fig. 5d).

Among the nested 97 solutes absent from all six supervised response sources, the corresponding contrast is 2.138 to 1.536 (paired change −0.603; interval, −0.863 to −0.359). The strict subset’s median absolute error moves in the opposite direction, from 1.102 to 1.244; the mean improvement is therefore not a gain for every molecule or every error summary (Supplementary Table 10). Absolute errors are also much larger here than on ARROW-85.

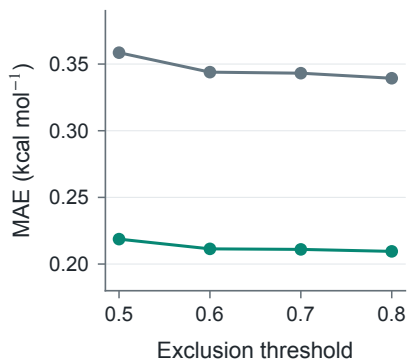
The largest absolute gain occurs where external molecules are least like the endpoint training pool (Fig. 5e). Among 35 solutes with nearest training-set Morgan similarity below 0.30, MAE is 4.027 kcal mol^{−1} from structural features and 2.501 with responses. This is an encouraging reduction in a difficult region, but the remaining error is still substantial. The full similarity-bin counts and estimates are shown in the figure.

Structure only ● SolvAI ●

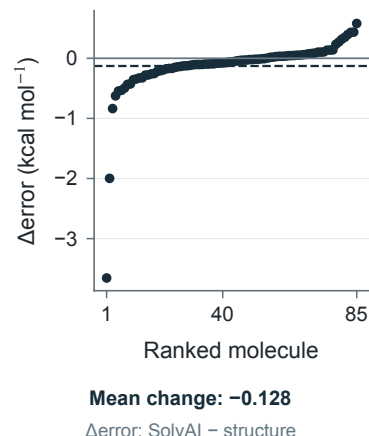
a Chemical groups



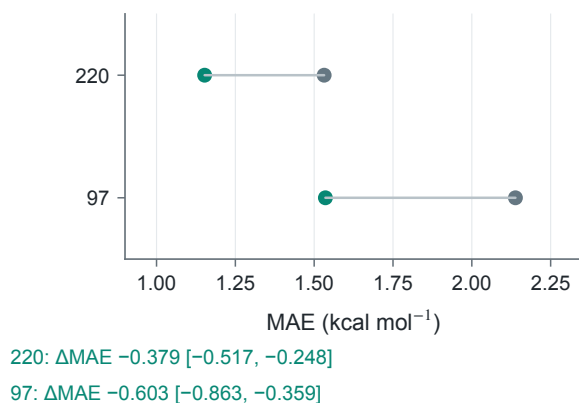
b Nearest neighbors



c No ARROW labels



d External matched comparison



e Relate error to chemical distance

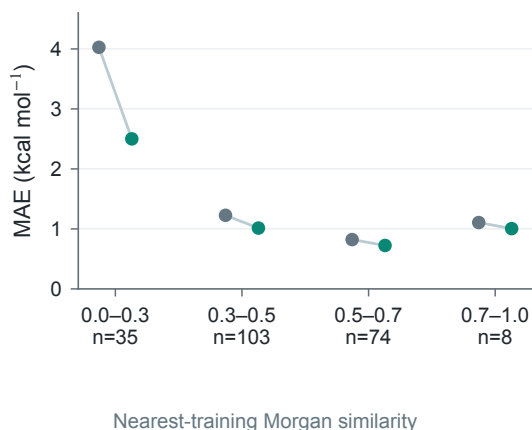


Figure 5: Response descriptors reduce error under chemical separation. **a**, Matched MAEs after removing every endpoint training label in the held-out molecular cluster, scaffold or functional family. **b**, Similarity exclusion removes every training row at or above the indicated Morgan similarity threshold to a test molecule. **c**, Molecule-level change in absolute error when no ARROW experimental label is used for training; negative values favor adding responses. **d**, Matched performance on the prospectively frozen external cohort and the subset also absent from every response-source table. Bracketed values below the plot are 95% intervals from 100,000 paired molecule resamples for the MAE difference, not intervals for the plotted MAEs. **e**, External errors in fixed nearest-training Morgan similarity bins, with all sample counts. The largest improvement occurs in the least similar bin, where absolute errors also remain largest. These bins describe the cohort, not calibrated applicability thresholds. All response surrogates use the standardized-equivalent exclusions described in Methods.

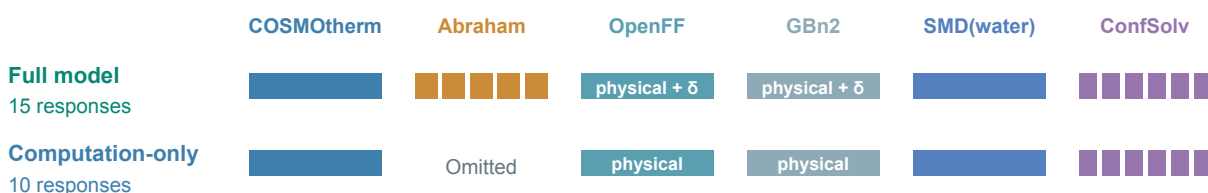
Source composition changes transfer behavior

The response set mixes computed quantities with empirical parameters and experimental residual corrections. To distinguish their roles, an additional control (Fig. 6) retains ten computed-source descriptors: COSMOtherm, uncorrected OpenFF and GBn2, SMD(water), and the six ConfSolv summaries. The predictor, experimental training labels and weights remain unchanged. On ARROW-85 this control reaches 0.234 kcal mol⁻¹; its improvement over structural features alone has a paired interval spanning zero (-0.191 to 0.022 kcal mol⁻¹). The full 15-descriptor model

is better than this control by $0.032 \text{ kcal mol}^{-1}$ (95% interval for full minus computation-only, -0.063 to -0.006).

The ordering reverses outside the reference chemistry. The ten-descriptor model reaches $1.095 \text{ kcal mol}^{-1}$ on the external 220 molecules, compared with 1.153 for the full model. The paired full-minus-computation-only difference is $0.057 \text{ kcal mol}^{-1}$ (95% interval, 0.023 to 0.093). On the strict 97-molecule subset, the corresponding errors are 1.422 and $1.536 \text{ kcal mol}^{-1}$, with a paired interval of 0.058 to 0.172 for the difference. Computed-source descriptors therefore reduce external error relative to structural features alone, while the full source composition is more effective on ARROW. This comparison identifies a dependence on source composition and evaluation chemistry, rather than a single source set that is uniformly preferable. Both fixed configurations are reported; external errors were not used to retune either model.

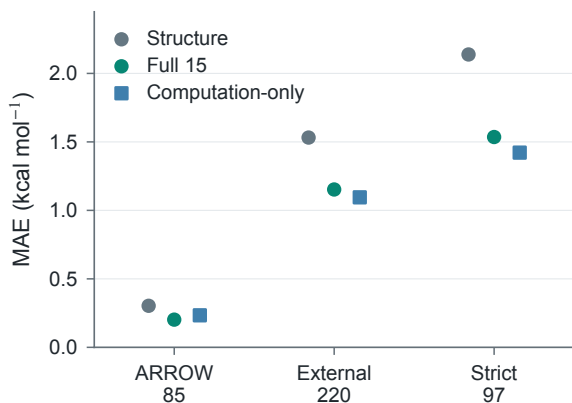
a Keep the learner fixed, change the response sources



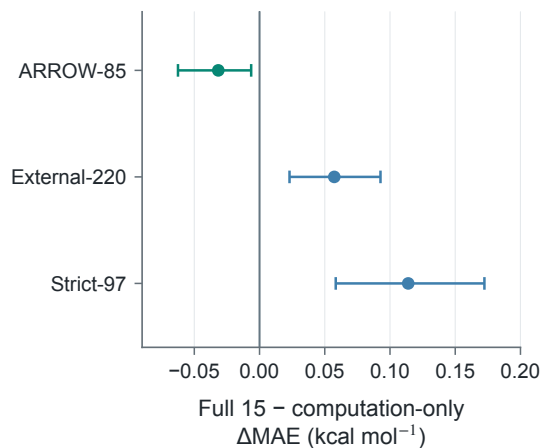
Remove five Abraham values and two residual corrections (δ). Retain the two physical energies.

Both models learn hydration from the same experimental labels.

b Compare absolute errors



c Compare the source sets directly



Negative favors full 15; positive favors computation-only

Figure 6: The preferred source composition changes with evaluation chemistry. **a**, The computation-only control removes five Abraham descriptors and both experimental residual corrections, δ , while retaining the underlying OpenFF and GBn2 physical-energy predictions. The resulting computation-only set contains ten descriptors. Both models retain the same structural features, experimental endpoint labels, weights and learner. Each tile denotes one descriptor, not its magnitude. **b**, Absolute MAEs for structural features, all 15 responses and the ten-response control on all three reported evaluation sets. **c**, Paired full-minus-computation-only MAE differences with 95% intervals from 100,000 molecule resamples. Negative values favor the full model on ARROW-85; positive values favor computation-only responses externally. The strict 97 are nested within the external 220. The external extension is retrospective and compares two fixed configurations without retuning on external errors. The reversal identifies a dependence on source composition and chemistry, not one uniformly optimal response set.

Changing source surrogates changes the transfer trade-off

The source mixture is only one design choice. Another is how accurately each source quantity can be inferred from structure, and whether improving that prediction helps hydration. In a retrospective challenge we held the 15 target definitions fixed and compared alternate learners for the four source families that had used trees. Source-only validation retained the released recipes for Abraham and ConfSolv but selected ridge-plus-neural-residual surrogates for OpenFF and GBn2 (Supplementary Method S22; Table 17). Their individual source targets were predicted more accurately, whereas the corrected OpenFF and GBn2 descriptors passed downstream showed smaller, unresolved changes. Better targetwise source scores did not automatically produce better hydration features.

Replacing only those two surrogate families made the trade-off visible in the hydration endpoint. With a protocol-matched ExtraTrees refit, ARROW-85 MAE moves from 0.203 to 0.238 kcal mol⁻¹, while External-220 improves from 1.192 to 1.075 and its source-disjoint Strict-97 subset from 1.595 to 1.352 (paired Strict-97 interval for the change, -0.358 to -0.137; Supplementary Table 18). A second endpoint has no clearly resolved gain, and the ExtraTrees size-held-out error rises. These results show that source-learner choice can favor one transfer setting without improving the others. The challenge does not replace the released model; source exposure and the full selection protocol are documented in Methods and Supplementary Method S22.

When response prediction is insufficient

Some plausible sources do not survive the two-stage route. In exploratory development screens, compact ConfSolv summaries helped more than the tested latent representations from the same source; additional force-field, pair-interaction and implicit-solvent features did not improve their matched endpoints (Supplementary Fig. 3). A useful source target must be relevant to hydration, predictable from structure and complementary to the information already supplied by other descriptors.

We also predicted coupling-path energy derivatives from short two-bead path-integral simulations (PIMD2). Although the surrogates captured broad trends, their component MAEs ranged from 1.27 to 5.20 kcal mol⁻¹. Adding the predicted response increased its separate exploratory endpoint error from 0.196 to 0.201 kcal mol⁻¹; no PIMD-derived coordinate enters the final model. The fuller hierarchy and direct-integration tests are in Supplementary Fig. 4 and Data 1. These outcomes test the learned descriptors in this setting, rather than rank the physical source methods themselves.

Testing transfer to larger molecules

We separate accuracy on labelled larger molecules from the shape of predictions for still larger, unlabelled series. For the first test, we held out the largest 15% within each endpoint-label source and refitted eight fixed configurations on 1,160 remaining labels. Of the 205 held-out molecules, 188 exceed the largest training molecule’s 17 heavy atoms. Their labels had been available in earlier random-fold development, so this is an exploratory test of endpoint size transfer with the source surrogates held fixed, not an independent test of the whole pipeline beyond all prior source exposure.

On this held-out split, TabPFN lowers MAE from 1.772 for ExtraTrees to 1.318 kcal mol⁻¹ (paired difference -0.453; 95% interval, -0.648 to -0.281; Fig. 7a). The responses also help TabPFN itself: its matched error without them is 1.959 (response-minus-control interval, -0.847 to -0.451). Several other neural endpoints improve on ExtraTrees here as well. They are useful alternatives for this split, although their separately selected input recipes mean the

cross-endpoint difference is not solely an architecture effect (Supplementary Table 15 and Data 6).

Figure 7 also follows capped glycine and alanine series without experimental peptide reference values. Those panels test how predictions change with chain length, not whether they are physically accurate.

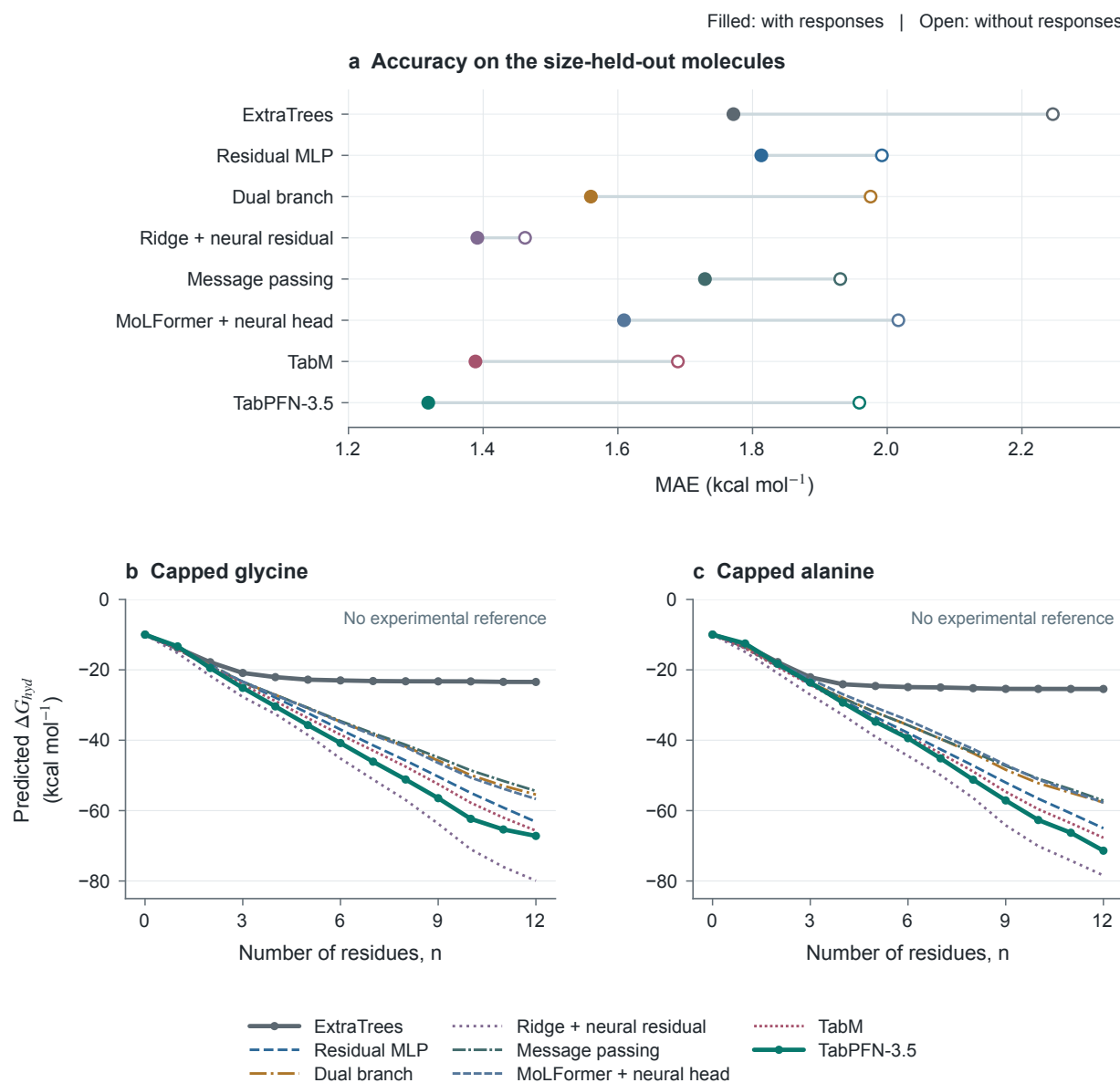


Figure 7: Labelled size transfer and unlabelled series answer different questions. **a**, MAE on the 205-molecule size holdout after refitting on the same 1,160 training labels. Filled and open marks denote models with and without responses. All predictions average three seeds. This diagnostic reuses development labels under a different split. **b,c**, Predictions along capped glycine and alanine series, using the full-data models with responses. Neural and TabPFN curves average five seeds; ExtraTrees averages three. These curves have no peptide reference values and therefore do not rank physical accuracy. The abscissa is the number of residues between the acetyl and N-methylamide caps; $n = 0$ is N-methylacetamide. Colors identify the same families in both panels.

In those unlabelled series, ExtraTrees predictions approach a plateau, while several neural endpoints and TabPFN keep changing as residues are added (Fig. 7b,c). That shape is less constrained by the fitted range, but it cannot be scored as more accurate without reference values. An alkane series makes the point from another direction: TabPFN is non-monotonic between C10

278 and C14, and most tested alkanes already occur in endpoint training. The individual predictions,
279 source-label audit and sensitivity to two alkane labels are in Supplementary Methods S20 and
280 Data 6.

281 Predictive uncertainty adds a final check on using this endpoint for new molecules. TabPFN’s
282 nominal 80% and 95% predictive intervals cover only 58.2% and 80.0% of External-220, respec-
283 tively, with similar undercoverage on Strict-97 (Supplementary Fig. 11 and Table 16). These
284 native intervals have not been calibrated on the external labels; their widths should not yet be
285 read as reliable per-molecule error guarantees.

286 Discussion

287 The value of SolvAI lies in changing when solvent information is used. Instead of resolving a new
288 solvent environment for every query, it learns molecular coordinates from earlier calculations
289 and measurements, then lets an experimental hydration endpoint decide how those coordinates
290 matter. Predicted physical descriptors and multi-fidelity transfer have precedents;^{8;9;18;19} the
291 test here is whether several heterogeneous solvent sources can be combined and reused across
292 new molecules and learners. The matched ExtraTrees contrast shows a gain beyond structural
293 features, and shuffling the response block among molecules removes it. Chemical-separation and
294 external-cohort tests show that part of this information remains useful beyond nearby endpoint
295 labels.

296 Two kinds of portability matter here. A response representation is reusable if different
297 hydration learners can extract value from it; a complete predictor transfers only if it remains
298 accurate in new chemistry. The first property is supported by the paired gains across several
299 endpoint families, including fits with pretrained molecular encodings. It does not follow from
300 ranking complete models by error: the lowest-error learner changes across cohorts, and its own
301 response ablation can be neutral or unfavorable. Keeping these questions apart shows what the
302 frozen surrogates contribute without attributing every model improvement to them.

303 The source-composition reversal shows why adding more teachers is not a monotonic route
304 to transfer. Empirical parameters and residual corrections help the full mixture on ARROW-
305 85, while the fixed computed-source subset has lower error on both external cohorts (Fig. 6).
306 Likewise, a more accurate surrogate for its own source target need not yield a better hydration
307 feature. These comparisons do not identify a single mechanism for the reversal; the source
308 distributions, prediction errors and endpoint use of each coordinate can all matter. They do
309 show that source selection must be evaluated on the intended deployment chemistry, with
310 independent labels held out from that choice.

311 The same distinction between promising behavior and verified accuracy governs larger
312 molecules. The labelled size split tests a different endpoint regime, but earlier development had
313 already seen its labels. Peptide and alkane curves show how predictions behave as structures
314 grow; without reference free energies, they cannot validate that behavior. The failed PIMD2
315 descriptors and undercovered predictive intervals further show that a physically motivated target
316 or a numerical uncertainty estimate is not automatically useful at deployment. The present
317 evidence is for neutral molecules near room temperature. It does not establish performance for
318 ions, other solvents, conformer-resolved energies or forces.

319 Independent larger-molecule and new-solvent references are the next decisive tests. They
320 would allow source learners to be judged by downstream utility as well as recovery of their own
321 targets, and would show how far solvent information learned once can replace a calculation
322 repeated for every new molecule.

Methods

Reference set and targets

The primary evaluation uses the 85 neutral organic solutes reported in the ARROW hydration analysis at 298 K.⁵ One water condition was retained per connectivity. The released benchmark table contains canonical structure, experimental hydration free energy, experimental uncertainty where available, classical ARROW, four-bead PIMD (PIMD4) and PIMD8 values, and all fold assignments. The exact reconstructed PIMD8 MAE is 0.205 kcal mol⁻¹; the source article reports approximately 0.20 kcal mol⁻¹.

Endpoint labels and features

The public training pool contains 1,280 neutral, benchmark-disjoint experimental hydration labels: 1,147 connectivity-resolved records from a FreeSolv/CombiSolv-Exp merge and 133 SoluteML dGsolvDB3 records supported by at least two source measurements.^{2;9;23} The ExtraTrees input has a 2,048-bit radius-2 Morgan fingerprint and 217 RDKit two-dimensional descriptors. Non-finite values are handled by a median imputer fitted inside each endpoint fit.

Response surrogates

Fifteen response descriptors are generated by six surrogate families (Supplementary Table 1). CheMeleon-initialized D-MPNNs predict two source families: CombiSolv-QM/COSMOtherm and MolSolv SMD(water).^{35;36} ExtraTrees models predict five Abraham axes and the physical and residual components of corrected OpenFF and GBn2 responses. LightGBM models predict six ConfSolv conformer and response-distribution summaries. The latter are derived from the H₂O subset of ConfSolv. The final coordinate list, target units, source sizes, transformations and inference calculations are given in Supplementary Table 1. PIMD-derived targets are absent from this list.

Retrospective source-surrogate architecture challenge

We separately compared five fixed learner candidates for each of four previously tree-based source families: the released tree recipe, an MLP using binary Morgan bits, an MLP using count features, count-aware ridge and count-aware ridge plus a bounded neural residual. The two D-MPNN source families, all source target definitions and the endpoint labels remained unchanged. Candidate choice used three source-development folds and a fixed 0.005 target-standardized-MAE tie margin favoring the simpler learner. A 20% source test, reserved before fitting, was opened only after choice was locked. The source-development choice of Abraham’s tree was retained despite the lower test point estimate of an unselected hybrid. Full candidates, preprocessing, targetwise and summed-source diagnostics appear in Supplementary Method S22 and Table 17.

Endpoint comparisons used three source conditions: the released response matrix; a protocol-matched refit of the four tree families; and that refit with only the preselected OpenFF and GBn2 hybrids substituted. Every exact endpoint molecule appearing in a source table received source out-of-fold predictions; none of the ARROW-85 or Strict-97 identities appeared in any supervised source table. ExtraTrees and ridge-plus-neural-residual endpoints were refitted with their own locked recipes, identical labels, folds and seeds across source conditions. External-220 includes source-exposed molecules and is secondary to Strict-97 for this source-stage challenge. Its paired intervals use 10,000 molecule resamples and are descriptive, not adjusted for model

365 exploration. The 205-molecule size refit and unlabelled alkane/peptide series are diagnostics,
366 not independent validation of extensivity (Supplementary Method S22 and Table 18).

367 Matched ExtraTrees endpoint

368 For the ExtraTrees endpoint, an ensemble of three `ExtraTreesRegressor` pipelines³⁰ is used,
369 each with 360 trees, maximum feature fraction 0.7, minimum leaf size 2, no bootstrap and
370 random seeds 11, 29 and 47. Predictions are averaged. Public endpoint rows have sample weight
371 1 and available ARROW outer-training rows weight 3. The matched structure-only model is
372 identical except that the 15 response columns are removed.

373 Identity standardization

374 RDKit 2026.03.5 was used to generate canonical isomeric SMILES, full InChIKeys and connec-
375 tivity blocks.³⁷ The confirmatory audit additionally applied MolStandardize cleanup, largest-
376 fragment selection, uncharging and canonical tautomer enumeration. Exact and standardized
377 benchmark equivalents were removed from every supervised external source. The audit found no
378 such match in the endpoint pool. It identified 2 CombiSolv-QM, 32 MolSolv and 22 ConfSolv
379 source records beyond the original exact-connectivity exclusion. The affected surrogates were
380 refitted with all remaining molecules held in their original train, validation or test split. Full
381 pair lists, identifiers and split-verification hashes are released.

382 Confirmatory evaluation

383 Confirmatory analyses followed a prespecified protocol archived in Git. Documented implemen-
384 tation amendments preceded the corrected runs (Supplementary Methods S9). The primary
385 partition is the existing five-fold assignment on the 85 molecules. Five complete shuffled K-fold
386 partitions use seeds 314159, 271828, 161803, 141421 and 173205. In each outer fold, all 1,280
387 public labels and the available ARROW outer-training labels are fitted; the test target is absent.
388 Feature blocks, ExtraTrees hyperparameters, sample weights and model seeds are identical
389 across all comparisons.

390 A separately prespecified one-factor sensitivity repeated the fixed folds after changing only
391 the available ARROW outer-training weight from 3 to 1. It retained all source rows, surrogates,
392 features, hyperparameters and seeds. Its paired interval used the same 100,000-resample
393 convention.

394 For the negative control, the 15 response columns are permuted as one block, thereby
395 preserving their joint distribution while destroying molecule alignment. Public, outer-training
396 and outer-test response rows are permuted separately within each fold. Five permutation seeds
397 (88001–88005) are evaluated for the fixed partition and each of the five repeats.

398 Chemical separation

399 Separation is applied to the complete endpoint-supervised pool. Functional families are assigned
400 by a deterministic SMARTS precedence; five outer folds are formed by group. Bemis–Murcko
401 scaffolds are canonicalized, with acyclic structures grouped by primary family. Taylor–Butina
402 clusters use radius-2, 2,048-bit Morgan fingerprints at Tanimoto similarity 0.70. For each outer
403 test fold, every public or ARROW training row sharing a held-out group is removed. A separate
404 nearest-neighbor analysis removes every training row with similarity at or above 0.50, 0.60,
405 0.70 or 0.80 to any test molecule. No threshold was selected by performance. Exact SMARTS,
406 assignments, training counts and maximum retained similarities are released in Supplementary
407 Data 3 and the accompanying `results/confirmatory/` tables.

Zero-ARROW transfer

The frozen endpoint configuration was fitted once to the 1,280 public experimental labels without any ARROW experimental value. Both matched structure-only and full response-descriptor models then predicted all 85 ARROW molecules. This analysis measures transfer without reference-set endpoint adaptation; it is not described as an independent external test because model development used the reference chemistry.

External validation

The candidate cohort was frozen for a separate solvation-model validation study, before that study’s model predictions, using original-measurement records from Sander’s Henry-law compilation v5.0.0.³⁴ The intrinsic Henry solubility constant H_s^{cp} is the aqueous concentration divided by the gas partial pressure. Values at 298.15 K were converted using $K_c = H_s^{cp}RT$ and $\Delta G_{\text{hyd}} = -RT \ln K_c$, corresponding to 1 M ideal gas and 1 M ideal-dilute aqueous standard states. Here K_c is the dimensionless aqueous-to-gas concentration ratio, R the gas constant and T the temperature. Qualification preceded SolvAI evaluation and required one connected neutral solute, supported elements, compatible target metadata and no exact, fragment-parent, uncharged-parent or canonical-tautomer equivalent in the 1,280 public or 85 ARROW endpoint labels. One previously documented registry conflict was excluded, leaving 220 molecules. A strict 97-molecule subset additionally excludes any identity-equivalent occurrence in all six supervised response-source tables.

For the matched ExtraTrees comparison, deployment-style fitting used the same 1,280 public plus all 85 ARROW labels, weights, preprocessing, architecture and seeds; only the 15 response columns differ. External-cohort labels were not used for fitting, surrogate or feature selection, thresholding or model choice in that comparison. The subsequent architecture study reuses this already examined cohort and is therefore retrospective, even though its final models do not fit these labels. The primary paired analysis and bootstrap convention match the confirmatory analysis. Complete eligibility decisions, source-exposure matches and molecule-level predictions are released as Supplementary Data 5. Because the compilation is external and molecular identities are disjoint, we call this an external molecule-disjoint cohort; we do not claim that all historical measurement provenance is independent.

Additional molecular-model comparisons

The graph and pretrained-representation comparisons followed a fixed protocol specified before their fits. These are retrospective extensions on the existing evaluation sets, not a second prospective validation. The same public labels, ARROW folds and sample weights are used for every model. Two direct predictors map molecular graphs to experimental hydration free energy: a D-MPNN trained from random initialization and a CheMeleon-initialized D-MPNN with its encoder fine-tuned.^{35;36} For each outer training set and seed, 15% of each label source is held out internally to select the epoch count by validation MAE, with an 80-epoch cap and patience 12. The model is then initialized afresh from the same random or pretrained starting weights and trained on the complete outer training set for that count. Only this refitted model predicts the outer test set. Three seeds (11, 29, 47) are averaged. Supplementary Fig. 6 illustrates the selection and refit protocol; Supplementary Methods S14 gives the complete configurations.

Two further comparisons use frozen pretrained representations: MolFormer encodes a SMILES sequence, and Uni-Mol encodes an RDKit-generated conformer.^{29;38} Each is concatenated with the 2,265 structural features and fitted with the same ExtraTrees ensemble as the matched control. We repeat the fit after adding the 15 response descriptors. This paired design tests whether the responses add value to pretrained features without changing the downstream learner.

454 The pretrained encoders receive no hydration labels during these fits; their original pretraining
455 membership has not been audited for molecular disjointness. All model families and both
456 members of each paired comparison are reported, irrespective of their ranking.

457 These matched evaluations compare alternative molecular-learning approaches without con-
458 flating different data splits or training resources.³⁹ We do not compare the reported errors
459 directly with unfiltered FreeSolv leaderboard values: 80 ARROW identities occur in FreeSolv,
460 and the supervised response sources require their own exclusions. A conventional FreeSolv
461 benchmark would require fold-specific refitting of every affected response source. The external
462 97-molecule subset instead tests all six response sources with explicitly audited exclusions.

463 The computation-only source control was specified separately before its ARROW fits. After
464 that comparison, we extended the same ten-descriptor configuration to the external cohort
465 without retuning. Its OpenFF and GBn2 components were recovered from the packaged surro-
466 gates and checked against the frozen corrected-response sums before evaluation (Supplementary
467 Methods S17). This sequence distinguishes the retrospective source analysis from the original
468 prospective external evaluation.

469 **Broader endpoint search and numerical reproducibility**

470 The extended study retains the five original ARROW outer folds, the external 220 identities,
471 the strict subset and the 15 frozen response predictions. For each outer training pool, random
472 inner splits hold out 15% of each source. The selection score is the equally weighted mean of
473 public-source and ARROW-source validation MAE, not an outer-test score. Neural fits use
474 source weights 1 and 3, respectively. Candidate screening is followed by repeated inner validation,
475 longer-training challenges up to 1,200 epochs and a separate width challenge. Each family has a
476 locked configuration and training schedule for each outer partition. Final fits restart from their
477 specified initialization, restore the inner-validation rows to training, and average seeds 11, 29,
478 47, 71 and 101. Matched no-response fits use the same recipe and schedule. The ExtraTrees
479 and frozen-encoder/tree controls use seeds 11, 29 and 47; common-three-seed comparisons are
480 also reported. Supplementary Methods S18 specifies the search and its limits.

481 MoLFormer representations are generated with the encoder’s deterministic evaluation option in
482 addition to evaluation mode. An audit found that evaluation mode alone left stochastic attention
483 projections active in the earlier cache. We regenerated all study and stress representations
484 from the same checkpoint and repeated both the neural-head and fixed-tree comparisons. Only
485 these corrected MoLFormer results enter the manuscript’s comparisons. Reload, repeated-query
486 and reordered-batch checks verify prediction replay; the packaged ExtraTrees endpoint does
487 not use MoLFormer and is unaffected. The discrepancy and corrected hashes are recorded in
488 Supplementary Methods S19 and Data 6.

489 TabPFN-3.5 uses the pinned full and Fast checkpoints with `tabpfn` 9.1.0, run locally under
490 the provider’s non-commercial license. We do not fine-tune the foundation model. Eight
491 candidates combine two checkpoints with four feature recipes; training-side challenges compare
492 eight versus 32 ensemble passes and single versus repeated ARROW context rows. Context
493 repetition represents source emphasis but is not mathematically equivalent to weighted neural
494 loss. Splits precede repetition. The pretrained model’s native preprocessing, including feature
495 selection, is retained; this is not an entirely tree-free preprocessing pipeline. Fitted contexts
496 contain training labels and are not redistributed.

497 **Size, series and predictive-uncertainty diagnostics**

498 The size diagnostic holds out the largest 15% within each source, with stable identity ordering
499 for ties. Every model uses the same 1,160/205 split and three seeds; no held-out label is used

in that refit. Configurations come from the locked external-fit recipe. Because the held-out labels were accessible in earlier random-fold development, this is an exploratory conditional test of size transfer. The source models and their source-level out-of-fold predictions are not rebuilt around this size split; exclusion from the endpoint refit does not imply exclusion from all source supervision (Supplementary Methods S5 and S20). The unlabelled grid comprises 30 linear alkanes and 13 members each of capped glycine and alanine series, including the common zero-residue molecule. Series identities, source overlap, fingerprint counts and all predictions are released.

For uncertainty evaluation, we transform each TabPFN distribution to the common raw-target scale and form an equally weighted mixture across five seeds. Central 80% and 95% intervals are quantiles of this mixture, not averages of seedwise quantiles. Empirical coverage is the fraction of external labels inside each interval. We also report widths and verify the mixture mean against stored point predictions. The external cohort is used only to evaluate these intervals, not to calibrate them. Supplementary Methods S20–S21 details these diagnostics.

Statistics

MAE in kcal mol⁻¹ is primary. Root mean square error (RMSE), median absolute error and the fraction of molecules improved are secondary. All comparisons are paired by molecule. Ninety-five percent percentile intervals use 100,000 molecule bootstrap resamples with seed 20260828. For repeated partitions, each molecule’s absolute error is first averaged across repeats before resampling. A predeclared comparison is positive when the complete paired interval for candidate-minus-control MAE is below zero, negative when above zero and neutral otherwise. We report all prespecified comparisons without multiplicity adjustment and do not treat the panel as a single confirmatory hypothesis test. No post-result feature selection is used within the confirmatory comparisons. The later architecture search uses training-side validation and a locked selection policy; its test intervals are descriptive, conditional on fitted models and not adjusted for the many architectures compared.

Released ExtraTrees deployment artifact

After evaluation was frozen, the response surrogates were fitted with standardized benchmark equivalents removed and the endpoint was refitted on the 1,280 public plus 85 ARROW labels. This all-data artifact is used for deployment and the endpoint-disjoint external evaluation; ARROW accuracy uses only the appropriate fold-specific fits. Inference parses the supplied SMILES with RDKit and converts it to canonical isomeric SMILES before computing descriptors and evaluating the six frozen response-surrogate artifacts; three endpoint regressors are averaged. This step is not a claim of invariance to tautomer, protonation or salt form. The runtime does not read an experimental value, family label, scaffold, fold identifier, trajectory or PIMD result. Software versions, hashes and a SMILES-only smoke test are reported in Supplementary Methods S12–S13 and Tables 7–8. On the release host, warm CPU inference took a median of 15.29 s for one molecule and 15.82 s for a batch of 32 (0.494 s per molecule). These measurements describe this artifact and hardware; they are not a same-hardware comparison with physical simulation or the other endpoint models.

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Author contributions

G.O. and M.L. conceived and designed the study. G.O. carried out the computational work. G.O., B.F. and M.L. interpreted the results and revised the manuscript.

Competing interests

The authors declare no competing interests.

Use of artificial intelligence

Generative AI tools assisted with this research and manuscript preparation, and the authors take full responsibility for the content.

Data availability

Benchmark records, held-out predictions, split assignments, audit tables and derived source manifests for the released model and endpoint-architecture study are provided in the SolvAI repository (Supplementary Data 1–6). Aggregate results of the retrospective source-surrogate challenge are included in Supplementary Tables 17–18 and the repository’s source-architecture evidence JSON. Complete fit-level and molecule-level records for that challenge are retained for the revised data release. Large third-party datasets are not redistributed where their size or license makes that inappropriate; URLs or DOIs, hashes, licenses and preprocessing instructions are supplied in `repro/DATA_PROVENANCE.md`.

Code availability

Source code for the released model, its packaged inference artifacts, confirmatory and endpoint-architecture analyses, figure generation, manuscript source and a locked environment are available at <https://github.com/Scientific-Computing-Lab/SolvAI>. A frozen-artifact path reproduces predictions, metrics, figures and PDFs without rerunning physical calculations for the released analyses. The endpoint-comparison directory additionally provides predictions, selection specifications and analysis code. TabPFN weights require the provider’s license; fitted contexts that contain non-redistributable training labels are not published. The packaged public inference command uses the ExtraTrees endpoint, and its runtime measurements do not describe the alternative endpoints. The source-surrogate challenge code and its full fit records are not part of the current public release and will accompany the revised data deposit.

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Supplementary Information

Supplementary Methods S1–S22; Supplementary Notes S1–S2; Supplementary Tables 1–18; Supplementary Figures 1–12; Supplementary Data 1–6.

Supplementary Methods

S1. Reference-set processing

The ARROW reference table was reconstructed from machine-readable project outputs and contains 85 unique neutral solutes in water at 298 K. One molecule-condition row was retained for each connectivity. The reference includes experimental hydration free energy and, where available in the source, uncertainty, classical ARROW, PIMD4 and PIMD8 values. Molecular names were not used for identity. Canonical isomeric SMILES, full InChIKey and the first InChIKey block were generated with RDKit 2026.03.5. The released benchmark table is `data/benchmark/arrow_solvation_master.parquet`.

The reconstructed ARROW/PIMD8 MAE is $0.204836 \text{ kcal mol}^{-1}$ and the classical ARROW MAE is $0.784648 \text{ kcal mol}^{-1}$. These values are direct errors against the same experimental column used for SolvAI evaluation; neither simulation value is an input to the final model.

S2. Public endpoint labels

The endpoint-label catalog initially contained 5,075 benchmark-disjoint connectivities. A frozen source-quality rule retained all 1,147 rows from an earlier FreeSolv/CombiSolv-Exp merge and 133 additional SoluteML dGsolvDB3 rows supported by at least two source measurements, giving 1,280 public labels. Duplicate structures were resolved at the connectivity level before this rule was applied. The source composition is listed in Supplementary Table 3. The raw merged tables are not redistributed where source terms are not explicit; URLs, versions, hashes and reconstruction commands are given in `repro/DATA_PROVENANCE.md`. CombiSolv-Exp originates from the collection of Vermeire and Green;⁹ the version used here was downloaded from the SolvBERT repository mirror.⁴⁰

For a random outer fold, these 1,280 rows were combined with the available ARROW outer-training molecules. Public rows received sample weight 1 and ARROW outer-training rows weight 3. The experimental value of an outer-test molecule was never used. In the zero-ARROW transfer experiment, only the 1,280 public labels were fitted and all 85 ARROW molecules were evaluated.

The weight of 3 was part of the prospective confirmatory protocol rather than a post-result choice. A separately prespecified one-factor sensitivity changed only the available ARROW outer-training weight to 1. Under this setting, matched structure-only and response-augmented ExtraTrees MAEs are 0.30977 and $0.20642 \text{ kcal mol}^{-1}$, respectively; the paired change is $-0.10335 \text{ kcal mol}^{-1}$ (95% interval, -0.21883 to -0.01995).

S3. Response-source processing

CombiSolv-QM. The water subset was deduplicated by solute connectivity and expressed in kcal mol^{-1} . After the original exact-connectivity exclusion, 3,961 structures were available. Conservative canonical-tautomer comparison found two additional benchmark equivalents; the confirmatory surrogate therefore used 3,959 rows.

SoluteML Abraham parameters. Five Abraham solute axes (E, S, A, B and L) were predicted from 8,098 benchmark-disjoint structures. These coordinates respectively describe excess molar refraction, dipolarity/polarizability, hydrogen-bond acidity, hydrogen-bond basicity and gas-to-hexadecane partition response. No standardized benchmark equivalent was found in the audited source.

OpenFF explicit-water response. The source contains 520 benchmark-disjoint absolute solvation-free-energy records from an OpenFF 2.3.0/NAGL/TIP3P protocol with 14 alchemical states. The records come from the Open Free Energy (OpenFE) alchemical benchmark resources release,⁴¹ using version 2.3.0 of the OpenFF Sage force-field family.²⁴ Separate ExtraTrees models predicted calculated hydration free energy and its experimental residual, defined as experimental hydration free energy minus the original calculated hydration free energy, not minus a surrogate prediction. Their sum is the retained corrected OpenFF coordinate.

GBn2 implicit response. The source contains 550 benchmark-disjoint GBn2 hydration calculations. GBn2 is the generalized Born model of Nguyen, Roe and Simmerling;²⁵ the calculation records were obtained from the dataset released by Katzberger and Riniker.⁴² Separate ExtraTrees models predicted the calculated value and its experimental residual. The sum is the retained corrected implicit-solvent coordinate. Its residual target uses the same experiment-minus-original-calculation definition.

MolSolv SMD(water). The source archive contains 1,729,545 M06-2X/6-31G* SMD calculations. The frozen selection retained all eligible neutral structures with at most 10 heavy atoms and a deterministic one-in-eight sample for structures with 11–18 heavy atoms. After deduplication and exact-connectivity filtering, 350,391 rows remained. Canonical tautomer standardization identified 32 additional benchmark-equivalent records. The confirmatory model used 350,359 rows while keeping every retained molecule in its original source split.

ConfSolv H₂O. The archive contains 5,392,567 H₂O conformer records for 43,693 solutes. We converted kJ mol⁻¹ to kcal mol⁻¹ and aggregated conformers into gas, water and hydration conformer corrections together with water-response distribution summaries. The broader quality filter produced 39,878 neutral connectivities. The selected surrogate target set had 17,851 rows with complete features and admissible energy range; fragment-parent, uncharging and tautomer standardization removed 22 additional benchmark-equivalent records, leaving 17,829.

S4. The 15 response descriptors

Supplementary Table 1 defines every retained coordinate. At inference, the table’s “surrogate” column is the calculation that runs; the expensive source method does not. The final vector contains one CombiSolv-QM response, five Abraham axes, one corrected OpenFF response, one corrected GBn2 response, one SMD(water) response and six ConfSolv summaries, for 15 coordinates in total.

Richer CombiSolv, OpenFF, GBn2 and ConfSolv targets were available and were evaluated during model development. Only the listed coordinates belong to the released stack. In particular, no PIMD2, PIMD4, PIMD8, $dH/d\lambda$ or NQE coordinate is present.

S5. Surrogate models

The CombiSolv-QM and MolSolv targets were fitted with Chemprop 2.2.4 directed-message-passing models initialized from CheMeleon. Both used a two-layer feed-forward head of width

256, dropout 0.1, initial/maximum/final learning rates of 3×10^{-5} , 10^{-4} and 10^{-5} , and PyTorch seed 20260826. CombiSolv-QM used RDKit two-dimensional descriptors, batch size 64, a 30-epoch cap and patience 7; MolSolv used the molecular graph, batch size 256, a 25-epoch cap and patience 6. The original frozen train/validation/test memberships were 80/10/10% for CombiSolv-QM and 90/5/5% for MolSolv. Standardized equivalents were removed from their existing split rather than resplitting the remaining source.

Abraham, OpenFF and GBn2 targets used ExtraTrees regressors on the same 2,265 structure features used by the endpoint. The selected leaf sizes and target-level OOF errors are recorded in the source metadata. ConfSolv used 217 RDKit descriptors and the 256 highest-variance Morgan bits. Each target was fitted with a LightGBM L1-regression model with 450 estimators, learning rate 0.04, 31 leaves, minimum child size 20, row subsampling 0.8, column subsampling 0.55 and L_2 regularization 0.1. Target-specific seeds were 20260826 plus the target index for validation and 1000 plus that index for the all-retained-source fit.

For OpenFF and GBn2, source records were matched to public endpoint candidates by connectivity. Public molecules represented in a source table received predictions from five-fold source-level out-of-fold fits for each physical and residual target; other public molecules and ARROW molecules received the corresponding full-source model predictions. Minimum leaf size was selected from 1, 2 and 3 using those same source folds. Thus the recorded source OOF errors are selection scores, not nested unbiased validation estimates. These source folds were fixed independently of the later endpoint partitions; they were not rebuilt for chemical- or size-held-out endpoint fits. Source-OOF feature generation therefore differs from excluding an entire downstream holdout from source supervision. The source-preparation scripts are archived under `repro/response_sources/`.

The recorded OpenFF physical/residual OOF MAEs are 0.617/0.531 kcal mol⁻¹; the corresponding GBn2 values are 0.669/0.626. These are errors of separate source components, not of their corrected sum. The standardized-exclusion ConfSolv validation contains 1,785 molecules and gives MAEs of 0.248, 0.267, 0.354, 0.448, 0.688 and 0.450 kcal mol⁻¹ for coordinates 10–15 in Supplementary Table 1. These heterogeneous source checks do not form a common independent fidelity benchmark for all 15 coordinates; the endpoint comparisons assess their predictive utility rather than presuming uniform recovery accuracy.

S6. Matched ExtraTrees implementation

The molecular representation comprises a 2,048-bit radius-2 Morgan fingerprint and 217 RDKit two-dimensional descriptors. The confirmatory endpoint appends the 15 response descriptors, giving 2,280 columns. A median imputer is fitted on each training partition and adds missingness indicators only where required. Each of three `ExtraTreesRegressor` models uses 360 trees, squared-error splitting, `max_features=0.7`, `min_samples_leaf=2`, `min_samples_split=2`, unlimited depth and no bootstrap; seeds are 11, 29 and 47. Predictions are averaged. The matched structural model removes only the 15 response columns.

After all evaluation was frozen, the release artifact was refitted on the 1,280 public labels and all 85 ARROW labels using the same weights and configuration. That refit supplies deployment predictions and the endpoint-disjoint external evaluation. Reported ARROW held-out scores use only the appropriate fold-specific fits.

S7. Validation partitions

The fixed primary result uses the released `fold_random` assignment. Five complete repeated five-fold partitions use shuffled K-fold seeds 314159, 271828, 161803, 141421 and 173205. All comparisons use the same fold, training rows and model seeds within a partition.

For functional-family separation, SMARTS rules were applied in the precedence order carboxylic acid, amide, ester, aldehyde, ketone, alcohol/phenol, ether, amine, thiol, sulfide/disulfide, aromatic heterocycle, carbocyclic aromatic, alkene/diene, saturated hydrocarbon and other. A test-fold family was removed from both public and ARROW endpoint training rows. Bemis–Murcko scaffold separation used canonical isomeric scaffold SMILES; acyclic molecules were keyed by primary family. Molecular clusters were constructed once on the combined endpoint pool with Taylor–Butina clustering at Morgan Tanimoto similarity 0.70. Nearest-neighbor analyses excluded training rows at similarity thresholds 0.50, 0.60, 0.70 and 0.80 to any outer-test molecule.

S8. Shuffled-descriptor control

All 15 response columns were permuted together to preserve their joint distribution. For each outer fold, public, ARROW outer-training and ARROW outer-test response rows were permuted separately. The experimental target was never used to define a permutation. Five base seeds (88001–88005) were used for the fixed partition and each repeated partition. This control changes molecule–response alignment while retaining feature count, response marginal distributions and inter-descriptor correlations.

S9. Statistical analysis

MAE is primary. RMSE, median absolute error and the fraction of molecules improved are secondary. Paired candidate-minus-reference differences were resampled at molecule level 100,000 times with seed 20260828. The 2.5th and 97.5th percentiles define the reported intervals. For the five repeated partitions, each molecule’s absolute error was first averaged over partitions before paired resampling. The prespecified protocol defined a positive comparison as a complete interval below zero. We report all prespecified comparisons without multiplicity adjustment and do not treat the panel as a single confirmatory hypothesis test. The protocol and its implementation amendments are archived in Git at `release/CONFIRMATORY_FREEZE.md`; the amendments preceded the corrected runs. For the additional molecular-model comparisons, the released `results/endpoint_models_20261006/corrected_original_metrics.csv` also provides the coefficient of determination (R^2) for each model and cohort.

S10. External validation

The original 221-molecule cohort was frozen for a separate solvation-model validation study before that study’s model predictions. It was derived from type-M original-measurement records in Sander’s Henry-law compilation v5.0.0. Intrinsic pure-water H_s^{cp} values (aqueous concentration divided by gas partial pressure), normalized to 298.15 K, were converted as $K_c = H_s^{cp}RT$ and $\Delta G_{\text{hyd}} = -RT \ln K_c$, where K_c is the dimensionless aqueous-to-gas concentration ratio. This produces the same 1 M gas-to-1 M aqueous standard-state convention used for the endpoint. The source values are printed to two significant figures.

SolvAI qualification was performed without reading SolvAI predictions or errors. All 221 rows passed the connected, neutral, supported-element and target-compatibility rules. One candidate was excluded because that study’s registry audit had documented a definite conflict between its registry name/CAS and supplied InChIKey structure. The remaining 220 molecules had no exact, fragment-parent, uncharged-parent or canonical-tautomer equivalent among the 1,280 public or 85 ARROW endpoint labels. Of these, 123 occurred in at least one supervised response-source table and 97 were absent from all six. The latter define the strict response-source-disjoint subset. The cohort hashes, all exclusions and all response-source matches were recorded in Git before endpoint evaluation.

Both endpoint models were then fitted deployment-style on the same 1,280 public plus 85 ARROW labels with weights 1 and 3. They used the same descriptors, imputation, three 360-tree ensembles and seeds; only the 15 response columns differed. No external-cohort label entered fitting or selection. On $N=220$, structure-only and response-augmented ExtraTrees MAEs were 1.53165 and 1.15255 kcal mol⁻¹ (paired change -0.37909 ; 95% interval, -0.51692 to -0.24817). On strict $N=97$, the corresponding values were 2.13830 and 1.53560 kcal mol⁻¹ (paired change -0.60271 ; interval, -0.86277 to -0.35858). The absolute errors delimit applicability and are not presented as PIMD8-level performance outside ARROW-85.

S11. Lambda-response probes

Short PIMD2 response fingerprints were available at $\lambda = 0.1, 0.5$ and 0.9 for 72, 74 and 73 molecules, respectively; 72 had complete three-point curves. The targets were the total solute-solvent $dH/d\lambda$ and its electrostatic and van der Waals components. A zero-valued polarization field was not interpreted. Structure-only heads were evaluated by OOF prediction, and their outputs were added to the fixed SMD(water) plus ConfSolv response baseline for this exploratory ablation (MAE 0.1959 kcal mol⁻¹). A separate direct-integration experiment used the predicted three-point curve with an affine calibration fitted inside each outer training fold. These targets were evaluated as candidate supervision and were not retained.

S12. Artifact verification

The public API accepts one or more SMILES strings. RDKit parses each input and emits canonical isomeric SMILES before feature generation. It does not standardize a query across tautomers, protonation states or salt forms. The runtime computes descriptors, evaluates the two D-MPNNs and four tree-artifact families, and passes the resulting 2,280-vector to three endpoint regressors. The artifact audit checks file hashes, feature count and names, software versions and two smoke predictions. It rejects features containing experimental target, PIMD, trajectory, probe, fold or scaffold information. The standardized-exclusion surrogate refit is identified in the model card and manifest. The ensemble spread is returned for diagnosis but is not a calibrated per-query applicability or reliability score.

S13. Runtime benchmarking

Runtime was measured end to end from input SMILES to prediction on the released artifact. The host used an Intel Core i7-4930K CPU; the installed NVIDIA RTX 3090 was not used for inference. Cold single-molecule latency, five warm single-molecule runs, three batch-32 runs and peak resident memory were recorded. Chemprop predictor startup dominates single-molecule latency. These measurements describe the released implementation, not the irreducible model cost. The ARROW/PIMD8 comparison uses the published 15-window protocol and is kept separate from same-host inference timing.

S14. Additional molecular-model comparisons

The additional molecular-model comparison uses the existing 1,280 public endpoint labels, 85 ARROW labels and 220 external test molecules. The protocol was fixed before the new fits, but these are retrospective comparisons on already examined datasets. The original fixed five-fold ARROW assignments are retained. For the external test, all 1,365 endpoint labels are used; the 97-molecule source-disjoint subset is fixed before these comparisons and is not a separate fit. Public labels have weight 1 and ARROW training labels weight 3 throughout. The wider

optimization study in S18 is a separate comparison, not a reinterpretation of these fixed 80-epoch fits.

The scratch D-MPNN has message dimension 300, three message-passing steps, mean aggregation, a regression head with two hidden layers of 300 units and dropout 0.1. The CheMeleon model initializes message passing from the published `chemeleon_mp.pt` weights, retains their encoder configuration and fine-tunes that encoder; it uses the same regression-head dimensions and dropout. Both optimize squared error using Adam and batch size 64, with constant learning rates 3×10^{-4} and 10^{-4} , respectively. Target scaling is fitted on the training subset only. For each outer partition and model seed, a deterministic 15% of each source is held out internally (split seed 20261002 + model seed). The epoch with lowest unweighted validation MAE is selected, with a cap of 80 epochs and patience 12. A model reinitialized from the same random or CheMeleon starting weights is then fitted on all outer-training labels for that many epochs, using a newly fitted training scaler. This restores the inner-validation molecules to the final training pool without exposing the outer test labels. Predictions from seeds 11, 29 and 47 are averaged. Every selected epoch, validation trace, training identity and held-out identity is retained. Supplementary Fig. 6 shows the order of inner selection, fresh refitting and held-out scoring.

MoLFormer uses the published `ibm-research/MoLFormer-XL-both-10pct` model at revision 7b12d946c181a37f6012b9dc3b002275de070314. The 768-dimensional representation is the attention-mask mean of the last hidden state, with maximum sequence length 256. Uni-Mol uses its 512-dimensional molecular representation from the `mol_pre_all_h_220816` checkpoint, including hydrogens. Its RDKit conformer generator uses seed 42. MoLFormer uses the corrected deterministic representations described in S19; Uni-Mol reuses its original label-free cache, with missing structures encoded by the same pipeline. Three of the 353 structures absent from the original Uni-Mol embedding cache used the package’s two-dimensional fallback after three-dimensional embedding failed; they are retained rather than excluded by prediction quality. Both pretrained representations are frozen during endpoint fitting. Molecular membership in their original pretraining datasets is not audited here.

For each pretrained representation, we fit the same three 360-tree ExtraTrees regressors on structural features plus that representation, then repeat with the 15 SolvAI response descriptors added. Imputation is fitted inside each training partition. All families, both feature conditions and all seeds are retained in the report. The comparison is limited to these fixed training configurations; it is not an exhaustive hyperparameter optimization or a claim to the best achievable performance of an architecture. Code and hashed inputs are provided with `scripts/run_journal_baselines.py`; independent aggregation is provided with `scripts/summarize_journal_baselines.py`.

S15. Conformational meaning of the response targets

The experimental endpoint is the equilibrium hydration free energy of a molecular species. It does not assign a separate target to each geometry of that species. The ConfSolv H₂O subset supplies conformer-resolved gas, water and solvation free energies, which are aggregated before surrogate training. For gas or water conformer energies G_i , define $g_i = G_i - \min_j G_j$. The conformer correction is $C = -RT \ln \sum_i \exp(-g_i/RT)$. Two response descriptors are the gas and water corrections. The third is the difference between the ensemble hydration estimate and the solvation energy of the lowest-gas-energy conformer. The remaining three are the population standard deviation of conformer solvation energies and the arithmetic mean and population standard deviation of $g_i^{\text{water}} - g_i^{\text{gas}}$. Source energies are converted from kJ mol⁻¹ to kcal mol⁻¹. These six summaries characterize the supplied conformer set; they do not establish complete conformational sampling or provide a geometry-specific prediction at inference. Supplementary

Fig. 7 illustrates these six summaries and the distinction between individual conformers and the molecular equilibrium endpoint.

S16. Error as a function of chemical similarity

For each external molecule, we use the maximum Tanimoto similarity of its radius-2, 2,048-bit Morgan fingerprint to any of the 1,365 endpoint-training molecules. Fixed descriptive bins are $[0, 0.3)$, $[0.3, 0.5)$, $[0.5, 0.7)$ and $[0.7, 1]$. We report counts, MAEs and paired differences within each occupied bin. These are descriptive strata of an existing cohort, not an independently calibrated applicability-domain threshold. In particular, similarity to endpoint training and membership in a response-source dataset are different quantities. The exact molecule-level similarities, predictions and bootstrap intervals are supplied in `results/journal_20261002/`.

S17. Computation-only response control

A separate source-composition analysis combines ten descriptors derived solely from computed source targets: COSMOtherm water free energy, uncorrected OpenFF and GBn2 free energies, SMD(water), and the six ConfSolv summaries. It excludes all five empirical Abraham parameters and both experimental residual corrections. The ten descriptors are added to the same 2,265 structural features, with unchanged five-fold ARROW evaluation, endpoint labels, sample weights, trees and three seeds. The comparison was specified before fitting this additional control. It measures whether computed-source descriptors alone improve this learner; it is distinct from the original three-descriptor computation-core ablation. After the ARROW comparison, we specified an extension of the same fixed ten-descriptor model to the external cohort. The OpenFF and GBn2 components are recovered directly from the original packaged surrogates. Before fitting, their corrected sums must reproduce the frozen external response cache to within 10^{-10} kcal mol⁻¹. All other response descriptors come from that cache. The external fit uses all 1,365 endpoint labels with the original weights and seeds. Both stages, exact predictions and input hashes are retained; the extension is explicitly retrospective.

S18. Endpoint architecture and optimization study

The study reuses the 1,280 public labels, 85 ARROW labels, 220 external identities and original five-fold ARROW assignments. Source-surrogate weights, targets and experimental labels are unchanged. It is retrospective, with adaptive development before a final selection lock; it is not a new prospective validation. Final outer predictions are made only after training-side selection is complete. The common selection score averages public-source and ARROW-source validation MAE equally. Three random inner splits each hold out 15% of each eligible source, using seeds $640601 + j$ for $j = 0, 1, 2$. Public training rows have loss weight 1 and ARROW rows weight 3. No external label participates in these fits or inner decisions.

The initial 52-candidate list covers residual MLPs, dual-branch networks, ridge-plus-neural residuals, pretrained-encoding heads and directed message passing. Tabular residual blocks apply layer normalization, two linear transforms with GELU activation and dropout, then add an identity skip. The dual architecture keeps a separate 32-unit response branch before concatenating it with the structural branch. The ridge model fits a training-only linear baseline with regularization 10 to the scaled RDKit and response columns, then trains a residual network on the remaining target. This is not a ridge model on seven selected responses with a tree residual. The graph variants use Chemprop bond message passing with either summed and mean atom embeddings plus atom count, or summed learned atomic outputs plus a global correction. These readouts carry size information but do not impose exact additivity of the

full free-energy predictor. Frozen-encoder neural heads use separate structure/encoding and response branches.

Candidate configurations vary widths 32–256, depth, dropout 0–0.2, learning rates from 3×10^{-4} to 3×10^{-3} , weight decay, squared or Huber loss, and plateau-reduced or cosine schedules. AdamW and batch size 128 are used. Ordinary neural fits have a 600-epoch cap. Median imputation, constant-column removal and numerical scaling are fitted inside each training split; response columns are retained even if constant. Binary Morgan bits remain unscaled. The skewed Ipc descriptor is transformed by $\log(1 + \max(\text{Ipc}, 0))$ or removed; BertzCT and Kappa1–3 use signed-log transforms. Physical counts, mass and surface descriptors are not clipped to the training range. Count-based fingerprints and later raw-count, log-count and no-fingerprint variants test whether loss of multiplicity in binary Morgan features limits size response.

Each candidate is screened on split 0 at seed 11. The best two per family proceed to three splits and seeds 11/29. The family winner is challenged with a 1,200-epoch cosine schedule on the same three splits at seed 47, against its ordinary schedule. The longer schedule is retained only if mean validation MAE decreases by more than $0.005 \text{ kcal mol}^{-1}$ and at least two of three pairs improve. A retained long schedule is completed at seeds 11/29, giving nine inner fits. Thus this challenge changes both duration and learning-rate trajectory, not duration alone. Fresh outer fits use the median selected epoch and the inner-derived schedule. A width boundary challenge expands selected 256-unit models to 512 units when a family selected the upper width in at least four of six partitions; graph message dimension is expanded as well. The same paired improvement rule determines retention.

TabM adds 12 configurations spanning three fingerprint treatments and four width, depth, dropout and numerical-embedding settings. Each model uses 32 parameter-sharing ensemble members; their mean is the prediction.³¹ The study uses TabM 0.0.3 and rtdl-num-embeddings 0.0.12. Representation refinements and the width extension are retained as explicit selection variants, not relabelled as separate architectures. Nine of 24 width challenges passed to nine-repeat completion. All 81 retained wider-fit optima occur before 95% of their training horizon. Longer schedules were selected in all six corrected-MoLFormer partitions, five dual and TabM partitions, three graph partitions, and two residual-MLP and ridge partitions. Supplementary Fig. 12 shows the paired duration and width diagnostics.

The search contains 1,812 inner evaluation records, including 174 TabPFN context evaluations. Gradient-based runs total 1,018,212 epochs; TabPFN’s 2,112 internal ensemble passes are not gradient epochs. Summed fitting time is 45.06 hours across overlapping workers, not elapsed time or GPU-hours. The defined search and boundary checks completed, but do not exhaust all architectures, regularization choices or training-data interventions. Full candidate lists and per-partition decisions are in Supplementary Data 6. Final comparisons retain every family, not only the global selector. Matched no-response models inherit the response model’s recipe, so the comparison does not estimate each structural-only architecture’s best achievable error.

S19. Pretrained representations and TabPFN configuration

The earlier MoLFormer cache used evaluation mode without enabling the model’s `deterministic_eval` option. Stochastic attention projections remained active. All 1,585 study representations and 56 stress-series representations were therefore regenerated from the same checkpoint with both settings enabled, and all valid MoLFormer endpoint comparisons were refitted. Identical repeated encoding gave zero change; changes in query order and batching produced representation differences below 3×10^{-5} . Fresh encoding through endpoint prediction agreed within $4.9 \times 10^{-6} \text{ kcal mol}^{-1}$. Corrected fixed-tree results replace the earlier cache-based values in Fig. 4 and Supplementary Tables 11–12. Legacy stochastic-cache branches remain labelled in the released

diagnostic data and are excluded from valid-model selection and the manuscript’s performance comparisons. Uni-Mol’s conformer pipeline and the packaged ExtraTrees endpoint are unaffected. Original encoder pretraining membership remains unaudited.

TabPFN-3.5 uses `tabPFN` 9.1.0 and full/Fast checkpoints from revision `06bf2ba35c80a92a3b9abb436b99cf49e7a0365e` of `Prior-Labs/tabPFN_3_5`.³³ Checkpoint SHA-256 values are recorded in Supplementary Data 6. This specific software release is distinguished from the earlier tabular foundation-model paper.³² We use its pretrained weights without fine-tuning. Inputs retain training-fitted native preprocessing, including feature subsampling/ranking and target transformations. The use of a transformer endpoint therefore does not mean that all preprocessing or the six source-surrogate families are neural.

Eight candidates combine full/Fast weights with four recipes: RDKit plus binary Morgan bits; RDKit plus raw Morgan counts; RDKit alone; or RDKit plus the corrected MoLFormer representation. Each has the 15 responses in its full condition. The best two screen candidates receive repeated inner evaluation. Training-side challenges compare one versus three copies of each eligible ARROW context row, and eight versus 32 ensemble passes, using the same greater-than-0.005 and two-of-three improvement rule. Context repetition occurs after splitting and does not create new independent observations or equal weighted gradient fitting. All selected recipes use the full checkpoint and three ARROW copies. The selected ARROW folds 0–4 use respectively raw counts/8 passes, no fingerprint/32, no fingerprint/8, raw counts/32, and MoLFormer without fingerprint/8. The external fit uses MoLFormer without fingerprint and eight passes. These are a per-partition selection policy, not one identical input recipe across folds.

The comprehensive policy selects the eligible option with lowest repeated inner score independently in each partition. It selects TabPFN in all six partitions. Five-seed final predictions use seeds 11, 29, 47, 71 and 101; controls remove only the responses. Common-three-seed means address the unequal ensemble size relative to the three-seed ExtraTrees fits. Every fitted-state replay and final identity/label audit passed. Licensed TabPFN weights and fitted contexts are retained privately; contexts embed training labels and are not public model-weight substitutes.

S20. Size holdout, homologue series and label influence

The size split ranks molecules by descending heavy-atom count within public and ARROW sources, breaking ties by identity. The largest 15% comprise 192 public and 13 ARROW molecules, leaving 1,160 training rows. Training heavy-atom counts range from 0 to 17 and holdout counts from 8 to 37; 188 holdout molecules exceed the global training maximum. Models use their locked external-fit recipes and three fresh seeds. ExtraTrees is refitted on these 1,160 rows rather than evaluated using release weights that have seen the held-out labels. No size score affects selection. Since the held-out labels participated in earlier random-fold development, intervals describe this exploratory split, not prospective selection-free performance on new molecules. Supplementary Table 15 gives all valid family results. The 13 ARROW molecules retain their original exclusion from supervised source tables. No analogous source-disjoint guarantee is made for the 192 public molecules: source models and source-OOF features remain fixed, with no size-specific source refitting or exposure audit. Accordingly, this is a conditional endpoint comparison, not a validation of fully source-unseen size extrapolation.

The unlabelled grid contains linear C1–C30 alkanes and Ac-(Gly)_{*n*}-NHMe and Ac-(Ala)_{*n*}-NHMe for $n = 0, \dots, 12$. The two zero-residue rows represent the same N-methylacetamide molecule, so 56 rows are not 56 unique structures. All positive-length peptides are absent from endpoint training; N-methylacetamide is present. The longest glycine and alanine examples contain 53 and 65 heavy atoms, respectively, beyond the maximum 37 in the original training pool. Binary Morgan fingerprints are identical across $n = 2, \dots, 12$ within either peptide

family and across C7–C30 alkanes. Count fingerprints, RDKit descriptors and some predicted responses still change, so the entire input does not freeze. An initial neural failure involved very large Ipc excursions; the present transformed-input study avoids that numerical failure without establishing physical accuracy.

For ExtraTrees, each leaf value is an average of eligible training labels and the ensemble output is an average of leaf values. Averaging each tree’s minimum and maximum leaf values gives an outer prediction bound, not necessarily a jointly attainable extremum. The released ensemble bound is $[-32.5945, 5.7565]$ kcal mol⁻¹. TabPFN’s mean is a convex combination of finitely many bin means on its fitted raw-target scale; averaging seedwise extrema gives the wider outer bound $[-658.4946, 649.9221]$. Its edge distributions can have unbounded tails without an unbounded mean. Changing this bound is not a proof of size extensivity.

Only C17, C29 and C30 are absent from endpoint training. C14 and C16 labels of 2.16 and 1.22 kcal mol⁻¹ are faithful rounded means of SoluteML’s linked CompSol records (C14: 68956/68957; C16: 67229/67230). The revised CompSol archive retains these binary-mixture records. Critical IUPAC–NIST evaluations identify disagreement in the underlying solubility measurements and do not recommend a best solubility value (https://srdata.nist.gov/solubility/sol_detail.aspx?sysID=38_354 and https://srdata.nist.gov/solubility/sol_detail.aspx?sysID=38_389). We did not convert an extrapolated trend into a replacement label. A same-recipe three-seed refit omitting only C14/C16 raises their predictions from 2.858/2.206 to 4.118/4.204 kcal mol⁻¹. This isolates substantial label influence but does not establish that the refitted predictions are correct. All principal architecture comparisons retain the original labels.

S21. Native predictive intervals

For the five final external TabPFN fits, each predictive distribution is expressed on the same raw kcal mol⁻¹ grid. Equal-weight mixture cumulative probabilities are inverted to obtain central 80% and 95% intervals, rather than averaging individual quantiles. Reconstructed means agree with stored predictions within 1.4×10^{-5} kcal mol⁻¹. The external 220 labels and strict 97 subset are used to assess, not calibrate, coverage. At 80%, 128/220 and 54/97 labels are covered; at 95%, 176/220 and 74/97 are covered. Supplementary Table 16 also reports interval widths. These undercoverage rates preclude describing the native intervals as calibrated reliability estimates. Seed dispersion and bootstrap intervals for mean errors are different quantities. No peptide coverage can be measured without reference free energies.

S22. Retrospective source-surrogate architecture challenge

This analysis asks whether the learner used to predict a fixed source target changes source recovery and downstream hydration performance. It was designed after the released-model and endpoint-architecture evaluations and is exploratory, not a second prospective validation. The four challenged families are SoluteML Abraham (8,098 source molecules), OpenFF (520), GBn2 (550) and ConfSolv (17,829). CombiSolv-QM/COSMOtherm and MolSolv/SMD(water) retain their released D-MPNN surrogates. Source target definitions, source rows, original ARROW partitions, experimental endpoint labels and the 15-coordinate ordering were fixed. OpenFF and GBn2 each have a calculated-energy target and an experiment-minus-original-calculation residual target; the coordinate supplied to the endpoint is the sum of the two predictions, not either target alone.

For each challenged family, a deterministic random 20% source test was reserved with seed 20261009. The remaining 80% were divided into three shuffled development folds with seed 20261010. Five candidates were specified before source-test access: the released ExtraTrees or

LightGBM architecture refitted under the development splits; a residual MLP using the 217 RDKit descriptors and binary Morgan bits; the same MLP using count-aware Morgan features; count-aware ridge regression; and that ridge fit plus a bounded neural residual. Neural fits used two width-128 residual blocks, dropout 0.1, AdamW at learning rate 0.001, batch size 128, a 400-epoch ceiling and patience 35. A deterministic 10% subset of each training fold selected the stopping epoch; the outer source-development fold did not. Median imputation and feature scaling were fitted on training rows only. Ridge used $\alpha = 100$ on scaled RDKit and log-count features. Neural targets were scaled within the training fold and missing targets were masked in a Huber loss. Final source fits used the median chosen development-fold epoch. The tree comparator preserved its released target-specific leaf settings, and ConfSolv’s variance-filtered bits were chosen within each training fold.

Selection minimized the equal-target average of validation MAE divided by the training standard deviation of each target, averaged over the three folds. A precommitted difference smaller than 0.005 in this normalized score favored the simpler candidate in the listed order. Thus selection used neither the independent source test nor hydration test labels. An initial OpenFF development screen used an incorrect tree comparator leaf size; it was invalidated before source-test access, preserved in the experiment archive and rerun with the released target-specific settings. The corrected screen is the one reported. The selected models were the released tree for Abraham, ridge plus neural residual for OpenFF and GBn2, and the released LightGBM tree for ConfSolv. Supplementary Table 17 reports every independent source-test score. These normalized errors compare candidate learners *within* a family; different source targets and units make a ranking across families inappropriate.

The selected hybrid’s target-averaged source-test score was lower than the tree score for OpenFF (0.2295 versus 0.2645) and GBn2 (0.2387 versus 0.2544). Abraham’s unselected hybrid happened to have the lowest source-test point estimate (0.0948 versus 0.0973 for the selected tree), but the test did not reverse the development-fold choice. ConfSolv’s selected LightGBM tree also had the lowest source-test point estimate (0.4930). A separate diagnostic held out the largest 15% by heavy-atom count within each source. The normalized target-average score changed from 0.9002 to 0.7538 for OpenFF and from 1.0682 to 0.7118 for GBn2. Those scores again average separate targets. On the independent random source test, the *summed* corrected coordinate instead changed from 0.660 to 0.651 kcal mol⁻¹ for OpenFF (selected-minus-tree 95% paired interval, -0.142 to 0.128) and from 0.718 to 0.683 for GBn2 (-0.204 to 0.117). On the size-separated source test, the corrected OpenFF coordinate worsened from 1.896 to 2.282 kcal mol⁻¹ (interval, -0.059 to 0.847), while GBn2 changed from 2.100 to 2.063 (interval, -0.503 to 0.441). These intervals include zero. Component gains need not add when errors in the two predicted terms are combined.

Source-endpoint identity overlap was audited by exact standardized molecule. Every endpoint molecule appearing in a source table received source out-of-fold predictions under that source condition. The cross-fit counts were 1,281 for Abraham (1,189 public and 92 external), 520 for OpenFF (all public), 550 for GBn2 (all public) and 151 for ConfSolv (139 public and 12 external). No ARROW-85 or Strict-97 molecule overlapped a supervised source table. This distinction is especially important for the OpenFF and GBn2 experimental-residual targets: they are not independent simulation-only supervision. External-220 has 123 source-exposed identities and is secondary to the strict source-disjoint 97 for the source-stage question.

Three response conditions were compared downstream: the original released response matrix; *source-tree-refit*, with all four challenged families refitted using their released tree recipes and the same cross-fitting protocol; and *selected-all*, which substitutes only the validation-selected OpenFF and GBn2 hybrids into that refit. The two unchanged D-MPNN families were checked for exact equality across conditions. Each endpoint used identical labels, outer folds, structural inputs, seed set and locked learner recipe between source-tree-refit and selected-all. This

produced 162 matched ExtraTrees fits and 54 ridge-plus-neural-residual fits. Original-response flexible predictions and released ExtraTrees results replayed their earlier values. Supplementary Table 18 gives the six full condition-head results, rather than only favorable substitutions. Paired differences use 10,000 molecule-level bootstrap resamples; they are descriptive and not adjusted for the architecture search. For the ExtraTrees head, selected-all minus source-tree-refit gives ARROW-85 +0.0348 (95% interval, -0.0009 to 0.0771), External-220 -0.1175 (-0.1765 to -0.0603) and Strict-97 -0.2428 (-0.3585 to -0.1371) kcal mol⁻¹. For the flexible head, the changes are +0.0038, -0.0041 and -0.0233; all three intervals include zero.

The separate endpoint-size diagnostic holds out 192 public and 13 ARROW labels under the earlier within-source size rule (Supplementary Method S20). It is not a strict global heavy-atom cutoff. Selected-all versus source-tree-refit changes ExtraTrees MAE by +0.1233 kcal mol⁻¹ (interval, 0.0551 to 0.1945) and the flexible endpoint by -0.0141 (-0.0419 to 0.0128). Unlabelled n-alkanes C2-C20 and Ac-(Gly/Ala)₀₋₆-NHMe were also scored. The selected source hybrids continue changing on the alkane series beyond C10, unlike the corresponding source trees, but the ExtraTrees endpoint remains bounded and nearly plateaus on the peptide series. The flexible endpoint’s peptide predictions continue changing; no experimental peptide free energies are available to establish which shape is accurate. Neither diagnostic proves physical extensivity or conformational sensitivity. The released source representation and public deployment artifact were not replaced as a result of this retrospective comparison.

Supplementary Notes

S1. Identity relationship to FreeSolv

Eighty of the 85 ARROW reference connectivities have a FreeSolv identity. This fact does not establish the provenance of every experimental value in the ARROW study. It instead determines the leakage rule: those connectivities were removed from all external supervised sources before endpoint or surrogate fitting. The frozen SolvAI artifact is therefore not evaluated over unfiltered FreeSolv and no generic FreeSolv state-of-the-art claim is made.

S2. Interpretation of alternative supervision

Non-improving features are not interpreted as evidence that their underlying physical methods are inaccurate. A response can fail downstream because it is poorly predicted from structure, redundant with existing coordinates, mismatched to the endpoint or supported by too few labels. Compact ConfSolv summaries outperform larger graph and feed-forward latent representations in the matched exploratory screen. Sparse PIMD2 response produces physically recognizable trends but not the precision required for hydration-free-energy integration. These observations motivate the three considerations stated in the main text: relevance, structural learnability and non-redundancy. The screens do not isolate their individual causal effects.

Supplementary Data

The first five numbered groups are supplied under `paper/supplementary_data/` in the repository and source-data archive. CSV files are the archival tables; XLSX files provide corresponding worksheets. The filenames below are relative to that directory.

Supplementary Data 1: experiment ledger.

`Supplementary_Data_1_experiment_ledger.csv` and the same stem with `.xlsx` record the evaluated scientific experiments, including non-retained branches.

Supplementary Data 2: molecule-level predictions.

`Supplementary_Data_2_molecule_predictions.csv` and `.xlsx` contain experimental values, ARROW/PIMD8 values and held-out SolvAI predictions.

Supplementary Data 3: validation assignments.

`Supplementary_Data_3_split_assignments.csv` contains molecular identities, family, scaffold and cluster assignments; the `.xlsx` file additionally contains repeated-partition folds. Primary folds are included in Supplementary Data 2. Complete chemical-exclusion details are supplied in `results/confirmatory/` alongside the analysis outputs.

Supplementary Data 4: response and endpoint provenance.

`Supplementary_Data_4_teacher_priors.csv` defines the 15 descriptors; `Supplementary_Data_4_teacher_sources.csv` and `Supplementary_Data_4_endpoint_sources.csv` describe their sources. `Supplementary_Data_4_teacher_manifests.xlsx` collects these tables and the identity audit. Source versions, licenses and hashes are documented in `repro/DATA_PROVENANCE.md` and the linked manifests. The artifact filenames retain their original identifiers.

Supplementary Data 5: external-cohort validation.

`Supplementary_Data_5_tier_a_predictions.csv`, `Supplementary_Data_5_tier_a_qualification.csv` and `Supplementary_Data_5_tier_a_external_validation.xlsx` provide matched predictions, eligibility decisions and response-source exposure.

Supplementary Data 6: endpoint architectures and size diagnostics.

`results/endpoint_models_20261006/` contains the model/ablation metrics, held-out and unlabelled-series predictions, paired comparisons, size aggregates, native predictive intervals, selection specification and publication manifest. `repro/endpoint_models/README.md` describes reproduction and restricted training-context dependencies. Legacy stochastic-encoding rows are explicitly marked and are excluded from valid comparisons. Corrected original-comparison tables replace the MoLFormer rows without overwriting the earlier source archive.

Supplementary Tables

Where used, boldface identifies the best displayed point estimate within the comparison specified for that table; it does not establish statistical superiority.

Supplementary Table 1 | SolvAI response descriptors

Index	Artifact name	Physical meaning
1	combisolv_qm	COSMOtherm water response
2	abraham_e	excess molar refraction E
3	abraham_s	dipolarity/polarizability S
4	abraham_a	hydrogen-bond acidity A
5	abraham_b	hydrogen-bond basicity B
6	abraham_l	gas-to-hexadecane partition L
7	openff_corrected	explicit-water alchemical response
8	gbn2_corrected	implicit-solvent hydration response
9	smd_water	SMD water response
10	conf_gas_corr	gas conformer correction
11	conf_solution_corr	water conformer correction
12	conf_hydration_corr	hydration conformer correction
13	conf_gsolv_sd	conformer solvation-energy spread
14	conf_response_mean	mean conformer solvent response
15	conf_response_sd	conformer response spread

Indices	Source	Surrogate	Training rows	Transformation
1	CombiSolv-QM	CheMeleon D-MPNN	3959	direct
2–6	SoluteML	ExtraTrees	8098	direct
7	OpenFF 2.3.0 ASFE	ExtraTrees	520	prediction + residual
8	GBn2	ExtraTrees	550	prediction + residual
9	MolSolv	CheMeleon D-MPNN	350359	direct
10–15	ConfSolv H2O	LightGBM	17829	direct

Units are kcal mol⁻¹ except for indices 2–6, which use their respective Abraham scales. Every descriptor is inferred from structure by its surrogate; no source simulation is run at inference. ASFE denotes absolute solvation free energy. The six conformer summaries are defined explicitly in Supplementary Methods S15.

Supplementary Table 2 | Source-specific confirmatory filtering

source	unit	before standardized exclusion	removed	retained	role
CombiSolv-QM	unique structures	3961	2	3959	COSMOtherm water
MolSolv	unique structures	350391	32	350359	SMD(water)
ConfSolv	model-usable connectivities	17851	22	17829	conformer response
Endpoint labels	experimental connectivities	1280	0	1280	hydration free energy

Supplementary Table 3 | Public endpoint-label sources

source group	selected records	selection rule
FreeSolv only	69	benchmark-disjoint connectivity
CombiSolv-Exp only	588	water records; benchmark-disjoint connectivity
FreeSolv + CombiSolv-Exp	490	one duplicate-resolved connectivity
SoluteML dGsolvDB3	133	at least two source measurements

Supplementary Table 4 | Complete repeated-partition values

MAE is in kcal mol⁻¹; each repeat is a complete five-fold partition. Boldface marks the lower MAE within each repeated partition.

Repeat	Split seed	Method	MAE
0	314159	Structure only	0.31780
0	314159	SolvAI	0.20788
1	271828	Structure only	0.31059
1	271828	SolvAI	0.21010
2	161803	Structure only	0.31309
2	161803	SolvAI	0.21217
3	141421	Structure only	0.31602
3	141421	SolvAI	0.20618
4	173205	Structure only	0.30727
4	173205	SolvAI	0.20053

Supplementary Table 5 | Global chemical-separation results

MAE is in kcal mol⁻¹. NN denotes the nearest-neighbor similarity exclusion threshold: endpoint-training rows at or above that threshold are removed. Boldface marks the lower MAE within each separation regime.

Separation	Method	N	MAE
Functional family	Structure only	85	1.262
Functional family	SolvAI	85	0.468
Scaffold	Structure only	85	0.728
Scaffold	SolvAI	85	0.376
Cluster (similarity 0.70)	Structure only	85	0.349
Cluster (similarity 0.70)	SolvAI	85	0.216
NN exclusion 0.50	Structure only	85	0.359
NN exclusion 0.50	SolvAI	85	0.219
NN exclusion 0.60	Structure only	85	0.344
NN exclusion 0.60	SolvAI	85	0.211
NN exclusion 0.70	Structure only	85	0.343
NN exclusion 0.70	SolvAI	85	0.211
NN exclusion 0.80	Structure only	85	0.339
NN exclusion 0.80	SolvAI	85	0.210

Supplementary Table 6 | Diagnostic family errors

Fixed-partition out-of-fold response-augmented ExtraTrees errors, grouped by diagnostic family; these are not family-held-out predictions. MAE is in kcal mol⁻¹.

Family	N	MAE
Acids	3	0.230
Alcohols	13	0.196
Aldehydes	3	0.117
Alkanes	13	0.234
Alkenes	5	0.085
Amides	4	0.483
Amines	5	0.138
Aromatics	13	0.313
Esters	4	0.055
Ethers	4	0.241
Heterorings	5	0.122
Ketones	5	0.160
Sulfides	4	0.115
Thiols	4	0.095

Supplementary Table 7 | Release-artifact hashes

file	sha256	bytes
MODEL_CARD.md	a20bd031f5b85ba4f8fea701fa74f3d71f092ce79abb0c19b517711ff8ecfbde	1617
abraham_teacher.joblib	8c8a6365e7162dbe6ae445ce4004928ed86b2669d3e7a70d3d054f143ca97554	87055391
combisolv_qm.pt	2950be1232e2f868136e3388cf139e671d4196922ece0d3c7043d9dadece8b40	37452324
confsolv_teacher.joblib	a71487e53d5085e897dad02e62e95562933bc47262145fbf44286aae04b70b0d	5729365
head.joblib	fdc88cbae31696f254d85eb427b3c53a4811f29b3290d605b4f1ea66ebb95790	29424285
implicit_teacher.joblib	484d5e0c1efa759c8e7acbfcbdd87e218b93ae239e7f35303fe4cd7e310910f8	36316890
model_card.json	0f78177a701320a5aad77c45304ea73d5ca4799fb91e4f4fd01d0bb83f8e1244	1745
molosolv_smd.pt	35601030f030af72a9e6990bf8003df9d481858b27994b3fc70c81d605d130be	37226928
openff_teacher.joblib	55b7d31a293dd61fe57f5505e015775d7d86f53dca69980d398843706d58bb9b	48211661

Supplementary Table 8 | Runtime measurements

Measurement	Value
cold single molecule	15.312 s
warm single molecule median	15.294 s
warm single molecule p95	15.322 s
batch 32 median	15.824 s
batch 32 per molecule	0.494 s

Supplementary Table 9 | Equal-weight ARROW sensitivity

All error measures are in kcal mol⁻¹. Boldface marks the lower value for each error measure in this matched pair.

Method	N	MAE	RMSE	Median absolute error
Matched structure-only	85	0.30977	0.59585	0.14972
Full SolvAI	85	0.20642	0.32627	0.11653

Supplementary Table 10 | External validation

All error measures are in kcal mol⁻¹; the strict 97 are a subset of the 220. Boldface marks the lower error separately for each cohort and measure.

Cohort	Method	N	MAE	RMSE	Median absolute error
Endpoint-disjoint	Matched structure-only	220	1.53165	2.30974	0.96182
Endpoint-disjoint	Full SolvAI	220	1.15255	1.57723	0.84508
Strict response-source-disjoint	Matched structure-only	97	2.13830	3.08051	1.10237
Strict response-source-disjoint	Full SolvAI	97	1.53560	1.98792	1.24392

Supplementary Table 11 | Molecular-model comparisons

All error measures are in kcal mol⁻¹. For MoLFormer and Uni-Mol, “+ responses” retains the structural descriptors and adds the 15 response descriptors; it does not replace the structural input. Boldface marks the lowest displayed error within each cohort and measure among the methods in this table only.

Set	Model	N	MAE	RMSE	Median AE
ARROW-85	CheMeleon (fine-tuned)	85	0.4967	0.7473	0.3251
ARROW-85	Computation-only + structure	85	0.2339	0.3936	0.1400
ARROW-85	D-MPNN (from scratch)	85	0.5590	0.8300	0.4194
ARROW-85	MoLFormer + responses	85	0.2342	0.3671	0.1252
ARROW-85	MoLFormer + structure	85	0.3481	0.5984	0.1673
ARROW-85	SolvAI	85	0.2022	0.3176	0.1169
ARROW-85	Structural descriptors	85	0.3033	0.5861	0.1592
ARROW-85	Uni-Mol + responses	85	0.2204	0.3465	0.1388
ARROW-85	Uni-Mol + structure	85	0.3344	0.5818	0.1860
External-220	CheMeleon (fine-tuned)	220	1.2716	1.8674	0.8423
External-220	Computation-only + structure	220	1.0953	1.4961	0.7768
External-220	D-MPNN (from scratch)	220	1.3013	1.6348	1.1078
External-220	MoLFormer + responses	220	1.1738	1.5743	0.9621
External-220	MoLFormer + structure	220	1.5939	2.3404	1.1811
External-220	SolvAI	220	1.1526	1.5772	0.8451
External-220	Structural descriptors	220	1.5316	2.3097	0.9618
External-220	Uni-Mol + responses	220	1.1930	1.6038	0.9377
External-220	Uni-Mol + structure	220	1.5777	2.3311	1.1301
Strict-97	CheMeleon (fine-tuned)	97	1.7102	2.3769	1.1040
Strict-97	Computation-only + structure	97	1.4216	1.8709	1.1012
Strict-97	D-MPNN (from scratch)	97	1.5584	1.9131	1.2295
Strict-97	MoLFormer + responses	97	1.5635	1.9776	1.3206
Strict-97	MoLFormer + structure	97	2.2247	3.1159	1.3580
Strict-97	SolvAI	97	1.5356	1.9879	1.2439
Strict-97	Structural descriptors	97	2.1383	3.0805	1.1024
Strict-97	Uni-Mol + responses	97	1.5889	2.0088	1.3365
Strict-97	Uni-Mol + structure	97	2.2053	3.0961	1.2184

Supplementary Table 12 | Paired molecular-model comparisons

Differences are candidate minus reference MAE in kcal mol⁻¹; negative values favor the candidate. Intervals use 100,000 molecule resamples and are descriptive, without multiplicity correction. The strict set is a subset of the external set. Model labels follow Supplementary Table 11; “+ responses” also retains structure.

Set	Contrast	Δ MAE	95% lower	95% upper
ARROW-85	SolvAI – Structural descriptors	−0.1011	−0.2151	−0.0203

Set	Contrast	Δ MAE	95% lower	95% upper
ARROW-85	SolvAI – Computation-only + structure	−0.0316	−0.0625	−0.0064
ARROW-85	SolvAI – D-MPNN (from scratch)	−0.3568	−0.4742	−0.2498
ARROW-85	SolvAI – CheMeleon (fine-tuned)	−0.2944	−0.3887	−0.2109
ARROW-85	SolvAI – MoLFormer + structure	−0.1459	−0.2493	−0.0600
ARROW-85	SolvAI – MoLFormer + responses	−0.0319	−0.0541	−0.0110
ARROW-85	SolvAI – Uni-Mol + structure	−0.1322	−0.2362	−0.0492
ARROW-85	SolvAI – Uni-Mol + responses	−0.0181	−0.0346	−0.0018
ARROW-85	MoLFormer + responses – MoLFormer + structure	−0.1139	−0.2095	−0.0322
ARROW-85	Uni-Mol + responses – Uni-Mol + structure	−0.1141	−0.2195	−0.0288
ARROW-85	Computation-only + structure – Structural descriptors	−0.0695	−0.1912	0.0223
External-220	SolvAI – Structural descriptors	−0.3791	−0.5169	−0.2482
External-220	SolvAI – Computation-only + structure	0.0573	0.0230	0.0927
External-220	SolvAI – D-MPNN (from scratch)	−0.1487	−0.2733	−0.0208
External-220	SolvAI – CheMeleon (fine-tuned)	−0.1190	−0.2165	−0.0228
External-220	SolvAI – MoLFormer + structure	−0.4414	−0.5814	−0.3077
External-220	SolvAI – MoLFormer + responses	−0.0212	−0.0433	0.0008
External-220	SolvAI – Uni-Mol + structure	−0.4251	−0.5649	−0.2928
External-220	SolvAI – Uni-Mol + responses	−0.0405	−0.0597	−0.0210
External-220	MoLFormer + responses – MoLFormer + structure	−0.4201	−0.5619	−0.2862
External-220	Uni-Mol + responses – Uni-Mol + structure	−0.3847	−0.5247	−0.2522
External-220	Computation-only + structure – Structural descriptors	−0.4364	−0.5823	−0.2973
Strict-97	SolvAI – Structural descriptors	−0.6027	−0.8628	−0.3586
Strict-97	SolvAI – Computation-only + structure	0.1140	0.0584	0.1724
Strict-97	SolvAI – D-MPNN (from scratch)	−0.0228	−0.2694	0.2313
Strict-97	SolvAI – CheMeleon (fine-tuned)	−0.1746	−0.3437	−0.0117
Strict-97	SolvAI – MoLFormer + structure	−0.6891	−0.9478	−0.4439

Set	Contrast	Δ MAE	95% lower	95% upper
Strict-97	SolvAI – MoLFormer + responses	–0.0279	–0.0640	0.0083
Strict-97	SolvAI – Uni-Mol + structure	–0.6697	–0.9251	–0.4290
Strict-97	SolvAI – Uni-Mol + responses	–0.0533	–0.0822	–0.0247
Strict-97	MoLFormer + responses – MoLFormer + structure	–0.6612	–0.9282	–0.4102
Strict-97	Uni-Mol + responses – Uni-Mol + structure	–0.6164	–0.8764	–0.3722
Strict-97	Computation-only + structure – Structural descriptors	–0.7167	–0.9938	–0.4540

Supplementary Table 13 | Extended endpoint comparisons

All errors are in kcal mol^{–1}. Full predictions average five seeds for neural and TabPFN fits and three for the tree controls. The last column recomputes MAE using only the three shared seeds. No-response controls inherit selected recipes. Count- and width-adaptive rows are selection policies, not additional architectures. The three reported cohorts are not independent: Strict-97 is nested within External-220. Boldface marks the lowest displayed error within each cohort and measure among all rows, including no-response fits and selection policies; it does not imply statistical superiority.

Set	Endpoint	Input	MAE	RMSE	Median	3-seed MAE
ARROW-85	ExtraTrees	No response	0.3033	0.5861	0.1592	0.3033
ARROW-85	ExtraTrees	Response	0.2022	0.3176	0.1169	0.2022
ARROW-85	Residual MLP	No response	0.3581	0.6285	0.2086	0.3592
ARROW-85	Residual MLP	Response	0.3403	0.5384	0.2039	0.3308
ARROW-85	Dual branch	No response	0.3444	0.6234	0.1522	0.3478
ARROW-85	Dual branch	Response	0.2884	0.4574	0.1836	0.2893
ARROW-85	Ridge + neural residual	No response	0.3864	0.7521	0.2181	0.3822
ARROW-85	Ridge + neural residual	Response	0.3360	0.5268	0.2147	0.3343
ARROW-85	Message passing	No response	0.3222	0.6459	0.1412	0.3243
ARROW-85	Message passing	Response	0.2877	0.5178	0.1540	0.2956
ARROW-85	MoLFormer + neural head	No response	0.4647	0.7020	0.2792	0.4648
ARROW-85	MoLFormer + neural head	Response	0.2729	0.4241	0.1741	0.2883
ARROW-85	TabM	No response	0.3367	0.6415	0.1619	0.3415
ARROW-85	TabM	Response	0.2910	0.4608	0.1791	0.2901
ARROW-85	TabPFN-3.5	No response	0.2179	0.4384	0.1062	0.2166
ARROW-85	TabPFN-3.5	Response	0.1992	0.3916	0.0786	0.1965
ARROW-85	MoLFormer + ExtraTrees	No response	0.3481	0.5984	0.1673	0.3481
ARROW-85	MoLFormer + ExtraTrees	Response	0.2342	0.3671	0.1252	0.2342
ARROW-85	Count-adaptive selection	No response	0.3222	0.6459	0.1412	0.3243
ARROW-85	Count-adaptive selection	Response	0.2877	0.5178	0.1540	0.2956
ARROW-85	Width-adaptive selection	No response	0.3732	0.7292	0.1559	0.3847

Supplementary Table 13 (continued)

Set	Endpoint	Input	MAE	RMSE	Median	3-seed MAE
ARROW-85	Width-adaptive selection	Response	0.2879	0.5011	0.1455	0.2884
External-220	ExtraTrees	No response	1.5316	2.3097	0.9618	1.5316
External-220	ExtraTrees	Response	1.1526	1.5772	0.8451	1.1526
External-220	Residual MLP	No response	1.5246	2.1985	1.0809	1.5381
External-220	Residual MLP	Response	1.4281	2.0754	0.9788	1.4499
External-220	Dual branch	No response	1.5362	2.2491	1.0459	1.5402
External-220	Dual branch	Response	1.2339	1.7405	0.9392	1.2320
External-220	Ridge + neural residual	No response	1.1313	1.6178	0.7443	1.1272
External-220	Ridge + neural residual	Response	1.1723	1.6670	0.8471	1.1684
External-220	Message passing	No response	1.3110	1.8704	0.9396	1.2773
External-220	Message passing	Response	1.1678	1.6740	0.8554	1.1443
External-220	MoLFormer + neural head	No response	1.3325	1.8425	1.0735	1.3486
External-220	MoLFormer + neural head	Response	1.1113	1.5561	0.8664	1.1332
External-220	TabM	No response	1.2203	1.6453	0.9585	1.2177
External-220	TabM	Response	1.1317	1.5958	0.8627	1.1542
External-220	TabPFN-3.5	No response	1.1311	1.5627	0.8500	1.1534
External-220	TabPFN-3.5	Response	1.1339	1.6386	0.8355	1.1343
External-220	MoLFormer + Extra-Trees	No response	1.5939	2.3404	1.1811	1.5939
External-220	MoLFormer + Extra-Trees	Response	1.1738	1.5743	0.9621	1.1738
External-220	Count-adaptive selection	No response	1.1348	1.6092	0.7699	1.1349
External-220	Count-adaptive selection	Response	1.1479	1.6568	0.8179	1.1349
External-220	Width-adaptive selection	No response	1.3207	1.8695	0.9498	1.3250
External-220	Width-adaptive selection	Response	1.0904	1.5281	0.8664	1.0751

Supplementary Table 13 (continued)

Set	Endpoint	Input	MAE	RMSE	Median	3-seed MAE
Strict-97	ExtraTrees	No response	2.1383	3.0805	1.1024	2.1383
Strict-97	ExtraTrees	Response	1.5356	1.9879	1.2439	1.5356
Strict-97	Residual MLP	No response	2.1525	2.9344	1.5080	2.1652
Strict-97	Residual MLP	Response	2.0162	2.7645	1.4467	2.0605
Strict-97	Dual branch	No response	2.1889	3.0245	1.4705	2.1903
Strict-97	Dual branch	Response	1.6165	2.1756	1.3722	1.6141
Strict-97	Ridge + neural residual	No response	1.2941	1.7894	0.8337	1.2815
Strict-97	Ridge + neural residual	Response	1.4288	1.9521	0.9264	1.4289
Strict-97	Message passing	No response	1.7152	2.3425	1.2040	1.6307
Strict-97	Message passing	Response	1.5054	2.0904	1.1651	1.4489
Strict-97	MoLFormer + neural head	No response	1.6850	2.2703	1.2297	1.7110
Strict-97	MoLFormer + neural head	Response	1.4501	1.9473	1.0978	1.4830
Strict-97	TabM	No response	1.5655	2.0315	1.2785	1.5586
Strict-97	TabM	Response	1.4540	1.9517	1.1925	1.5056
Strict-97	TabPFN-3.5	No response	1.3467	1.8596	0.9823	1.3926
Strict-97	TabPFN-3.5	Response	1.4932	2.0476	1.0358	1.4854
Strict-97	MoLFormer + Extra-Trees	No response	2.2247	3.1159	1.3580	2.2247
Strict-97	MoLFormer + Extra-Trees	Response	1.5635	1.9776	1.3206	1.5635
Strict-97	Count-adaptive selection	No response	1.2974	1.7776	0.8305	1.2966
Strict-97	Count-adaptive selection	Response	1.3812	1.9387	0.8740	1.3502
Strict-97	Width-adaptive selection	No response	1.6271	2.2899	1.1027	1.6406
Strict-97	Width-adaptive selection	Response	1.3723	1.8633	1.0467	1.3222

Supplementary Table 14 | Matched response contrasts across endpoints

Within each endpoint, response-minus-no-response MAE and descriptive 95% paired molecule-bootstrap intervals, in kcal mol⁻¹; 100,000 resamples without multiplicity adjustment. Negative values favor adding responses. The candidate always includes responses; “No response” is its matched ablation. Boldface marks the largest within-endpoint reduction in each cohort, not the lowest absolute endpoint error.

Set	Endpoint	Δ MAE	Lower	Upper
ARROW-85	ExtraTrees	-0.1011	-0.2151	-0.0203
ARROW-85	Residual MLP	-0.0179	-0.0543	0.0127
ARROW-85	Dual branch	-0.0559	-0.1244	0.0066
ARROW-85	Ridge + neural residual	-0.0504	-0.1403	0.0263
ARROW-85	Message passing	-0.0345	-0.0997	0.0189
ARROW-85	MoLFormer + neural head	-0.1918	-0.2925	-0.1005
ARROW-85	TabM	-0.0457	-0.1138	0.0129
ARROW-85	TabPFN-3.5	-0.0187	-0.0463	0.0072

Supplementary Table 14 (continued)

Set	Endpoint	Δ MAE	Lower	Upper
ARROW-85	MoLFormer + ExtraTrees	-0.1139	-0.2095	-0.0322
ARROW-85	Count-adaptive selection	-0.0345	-0.0997	0.0189
ARROW-85	Width-adaptive selection	-0.0853	-0.1652	-0.0177
External-220	ExtraTrees	-0.3791	-0.5169	-0.2482
External-220	Residual MLP	-0.0965	-0.1372	-0.0575
External-220	Dual branch	-0.3023	-0.4151	-0.1960
External-220	Ridge + neural residual	0.0410	-0.0335	0.1180
External-220	Message passing	-0.1432	-0.2174	-0.0765
External-220	MoLFormer + neural head	-0.2212	-0.3096	-0.1343
External-220	TabM	-0.0886	-0.1324	-0.0441
External-220	TabPFN-3.5	0.0028	-0.0992	0.1122
External-220	MoLFormer + ExtraTrees	-0.4201	-0.5619	-0.2862
External-220	Count-adaptive selection	0.0132	-0.0642	0.0934
External-220	Width-adaptive selection	-0.2303	-0.3186	-0.1451
Strict-97	ExtraTrees	-0.6027	-0.8628	-0.3586
Strict-97	Residual MLP	-0.1362	-0.2132	-0.0625
Strict-97	Dual branch	-0.5724	-0.7824	-0.3750
Strict-97	Ridge + neural residual	0.1346	0.0068	0.2699
Strict-97	Message passing	-0.2098	-0.3468	-0.0889
Strict-97	MoLFormer + neural head	-0.2349	-0.3810	-0.0869
Strict-97	TabM	-0.1116	-0.1827	-0.0409
Strict-97	TabPFN-3.5	0.1465	-0.0470	0.3611
Strict-97	MoLFormer + ExtraTrees	-0.6612	-0.9282	-0.4102
Strict-97	Count-adaptive selection	0.0838	-0.0531	0.2277
Strict-97	Width-adaptive selection	-0.2548	-0.3934	-0.1184

Supplementary Table 15 | Exploratory size holdout

The same 205 held-out molecules and 1,160 training labels are used for every row. All fits average three seeds. Errors are in kcal mol⁻¹; the last column is each endpoint’s response-minus-no-response MAE, not a cross-endpoint ranking. Cross-endpoint paired intervals are in Supplementary Data 6, `size_diagnostic_comparisons.csv`. These development labels were available in earlier random-fold evaluation. Boldface marks the lowest displayed MAE in each absolute-error column and the largest response-associated reduction in the within-endpoint difference column.

Endpoint	Response MAE	No-response MAE	Δ MAE
ExtraTrees	1.7716	2.2459	-0.4743
Residual MLP	1.8130	1.9921	-0.1791
Dual branch	1.5600	1.9753	-0.4154
Ridge + neural residual	1.3913	1.4622	-0.0709
Message passing	1.7293	1.9303	-0.2010
MoLFormer + neural head	1.6091	2.0165	-0.4074
TabM	1.3886	1.6891	-0.3006
TabPFN-3.5	1.3185	1.9586	-0.6402
Count-adaptive selection	1.3369	1.4470	-0.1101
Width-adaptive selection	1.5215	1.9901	-0.4686

Supplementary Table 16 | Native TabPFN interval coverage

Intervals are central quantiles of the equal five-model predictive mixture. Widths are in kcal mol⁻¹; no external calibration was applied.

Set	Nominal	Covered	Observed	Mean width	Median width
External-220	80%	128/220	58.2%	2.605	1.634
External-220	95%	176/220	80.0%	4.605	2.860
Strict-97	80%	54/97	55.7%	3.539	2.728
Strict-97	95%	74/97	76.3%	6.200	4.678

Supplementary Table 17 | Source-surrogate learner challenge

Independent random-source-test scores are MAE divided by each target’s training standard deviation and averaged equally within each source family. Lower is better; scores are not comparable across families or with hydration MAE. Boldface marks the lowest displayed *test* point estimate in each row, not the selected model. Selection used only source-development folds and retained Abraham’s tree even though its unselected hybrid is bold here. The two released D-MPNN source families were not re-screened.

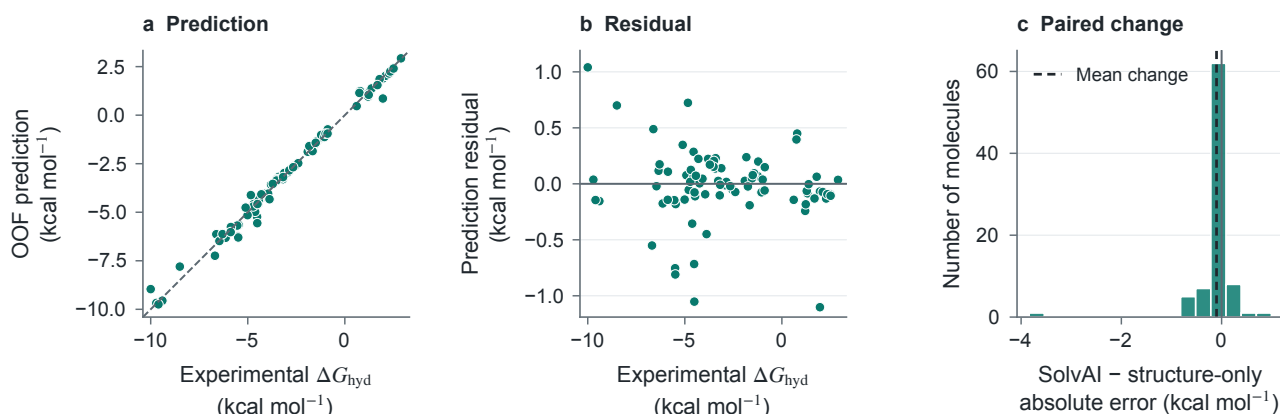
Source family	Tree	Binary MLP	Count MLP	Count ridge	Ridge + NN	Locked choice
Abraham	0.0973	0.1137	0.1121	0.1260	0.0948	Tree
OpenFF	0.2645	0.2585	0.2495	0.2742	0.2295	Ridge + NN
GBn2	0.2544	0.2770	0.2761	0.2548	0.2387	Ridge + NN
ConfSolv	0.4930	0.5147	0.5134	0.5699	0.5376	Tree

Supplementary Table 18 | Downstream source-model substitutions

Hydration MAE is in kcal mol⁻¹; Strict-97 is nested in External-220. The size diagnostic is a separate 205-molecule endpoint refit, not a global source-unseen size test. *Tree refit* uses a common cross-fitting protocol for the four challenged source families; *selected* substitutes the two source-validation-selected OpenFF and GBn2 hybrids. Comparisons between those two rows within the same endpoint head hold labels, folds, features and learner fixed. *Released* uses the original frozen response matrix and is retained as a second reference. Boldface marks the lowest displayed point estimate in each column, not a statistically established winner.

Endpoint head	Source condition	ARROW-85	External-220	Strict-97	Size-205
ExtraTrees	Released	0.2022	1.1526	1.5356	1.7716
ExtraTrees	Tree refit	0.2031	1.1922	1.5945	1.7483
ExtraTrees	Selected	0.2379	1.0747	1.3517	1.8716
Ridge + neural residual	Released	0.3343	1.1684	1.4289	1.3913
Ridge + neural residual	Tree refit	0.3298	1.2539	1.6400	1.3062
Ridge + neural residual	Selected	0.3336	1.2498	1.6166	1.2921

Supplementary Figures

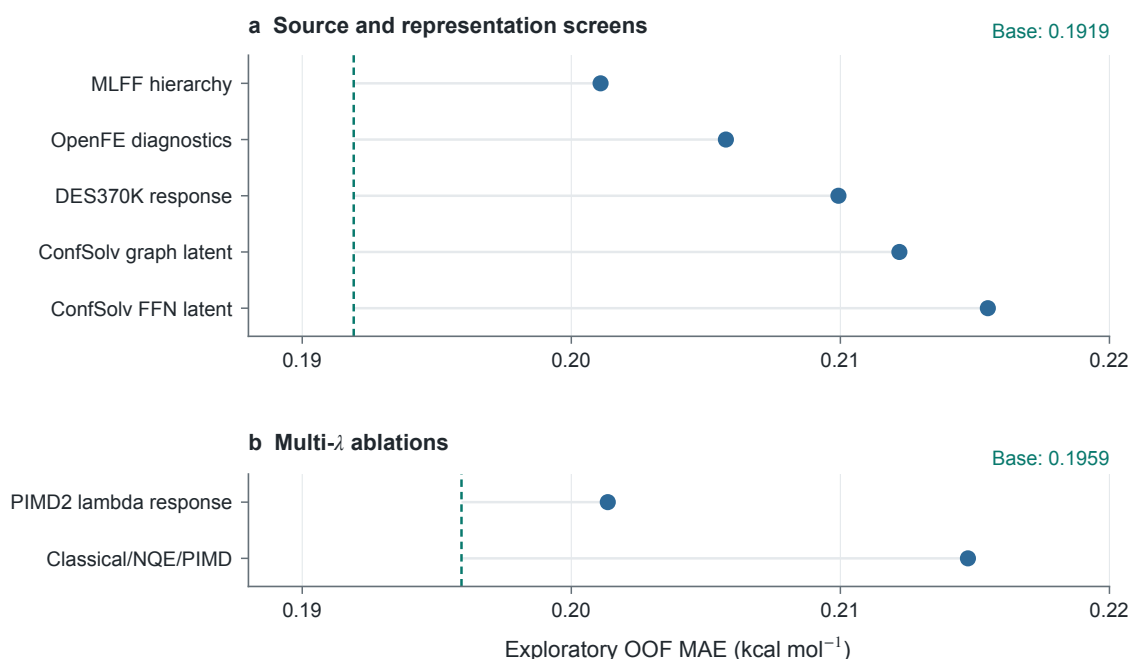


Supplementary Figure 1 | Molecule-level prediction and residuals. **a**, Confirmatory response-augmented ExtraTrees OOF prediction against experiment. **b**, Signed residual (prediction minus experiment) against experimental hydration free energy. **c**, Distribution of the paired absolute-error change, responses minus the matched structure-only endpoint; negative values favor responses.

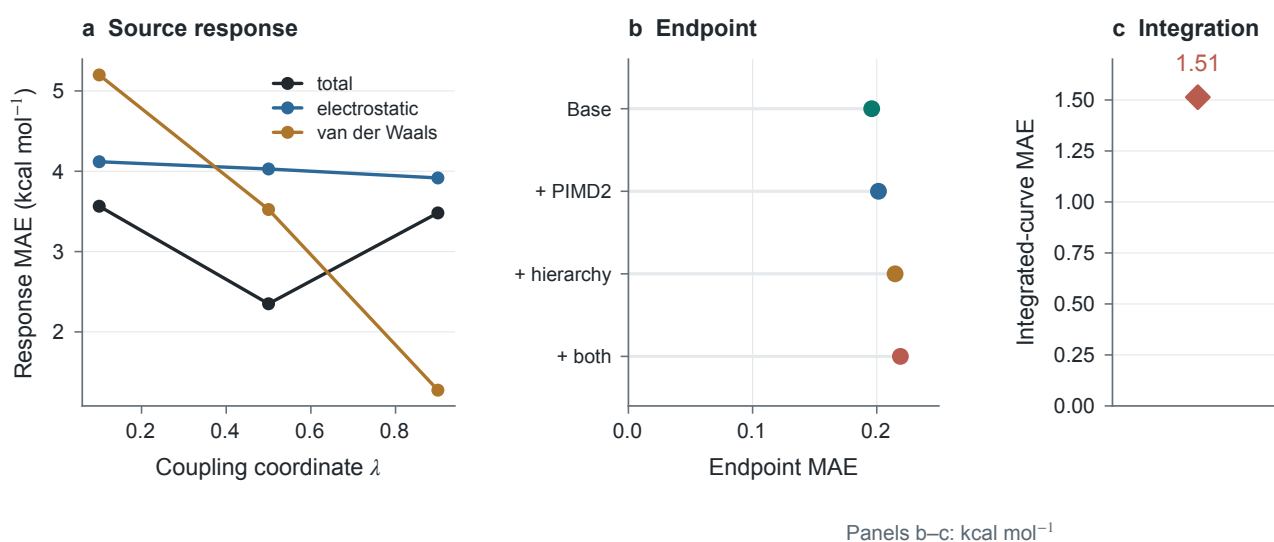
Source	Counting unit	Before	Removed	Retained
CombiSolv-QM	structures	3,961	2	3,959
MolSolv	unique structures	350,391	32	350,359
ConfSolv	usable connectivities	17,851	22	17,829
Endpoint labels	connectivities	1,280	0	1,280

Counts retain their source-specific units; they are not pooled.

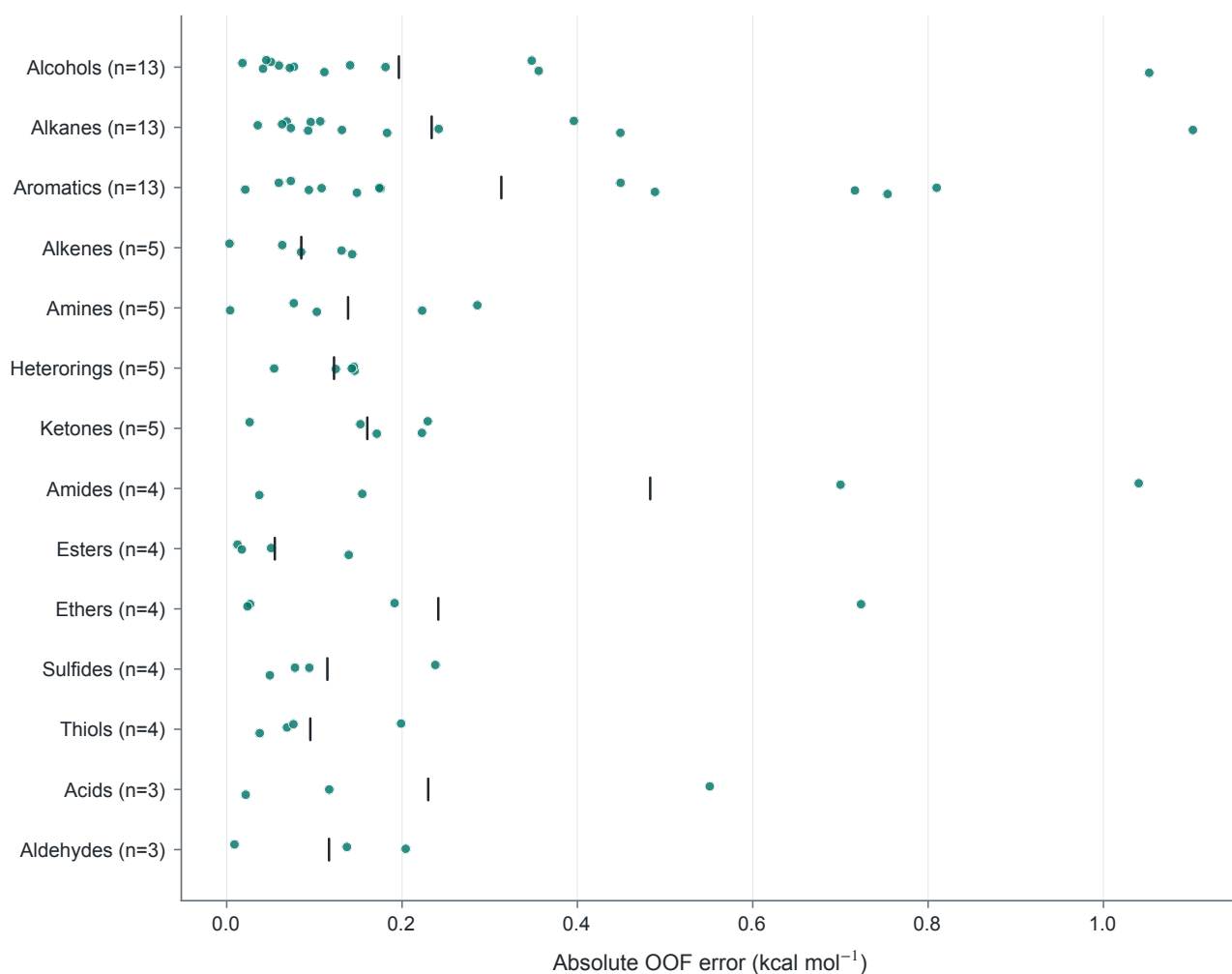
Supplementary Figure 2 | Source-specific provenance and standardized identity exclusion. Counts before exclusion, removed counts and retained counts are shown in their native source units. The audit removed standardized benchmark equivalents before the confirmatory surrogate refits; no endpoint experimental label was removed.



Supplementary Figure 3 | Alternative physical-supervision routes. Exploratory fixed screens of additional diagnostic, latent and hierarchy features. **a**, Source and representation screens use a one-seed SMD(water) plus ConfSolv development base (0.1919 kcal mol⁻¹), whereas the final 15-coordinate result averages three seeded fits. **b**, Multi- λ ablations use a separate fixed SMD(water) plus ConfSolv baseline (0.1959 kcal mol⁻¹). Each dashed line marks the corresponding control; gray segments connect that control to each screened alternative. These experiments were not used to select the confirmatory controls. OpenFE denotes the Open Free Energy toolkit; MLFF denotes machine-learning force field; DES370K denotes the roughly 370,000-geometry intermolecular-interaction dataset; FFN denotes feed-forward network.



Supplementary Figure 4 | Multi- λ response experiment. **a**, OOF response-head error at three coupling states. **b**, Matched endpoint consequences. **c**, Separately scaled error obtained by direct integration of the predicted curve with fold-local affine calibration (diamond).



Supplementary Figure 5 | Molecular errors by chemical family. Individual fixed-partition response-augmented ExtraTrees errors are grouped by diagnostic chemical family; ticks show family means and labels give sample sizes. Global separation results are shown in Fig. 5a,b.

a Keep test labels outside fitting and selection

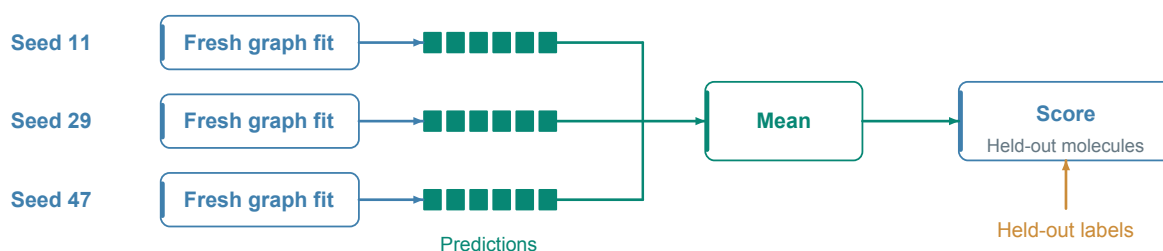
Outer-training labels



The 15% inner split is selected separately within each label source.

b Average independent seeds before scoring

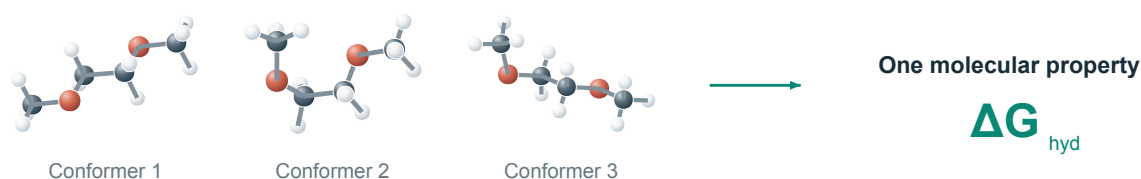
Repeat selection and refit for every seed



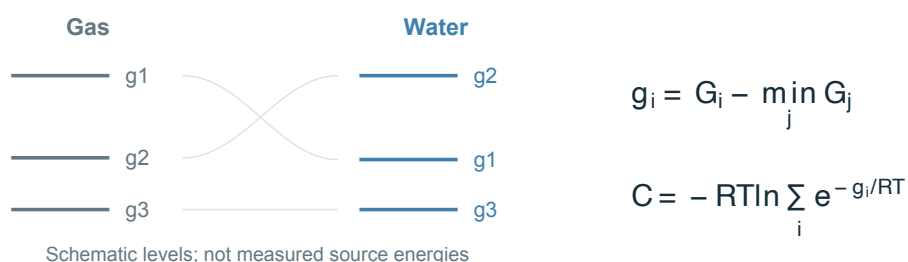
Five ARROW folds + one external fit; three seeds × two direct graph models = 36 runs.

Supplementary Figure 6 | Select graph-model duration without using held-out labels. **a**, Within each outer-training set and seed, a deterministic 15% of each label source is reserved for inner validation. The epoch with the lowest unweighted validation MAE is selected, subject to an 80-epoch cap and patience 12. A model reinitialized from its original random or CheMeleon starting weights is then refitted on all outer-training labels for that many epochs, with a newly fitted training scaler. **b**, Selection and refitting are repeated for seeds 11, 29 and 47. Their predictions are averaged before comparison with held-out labels. The five ARROW folds and one all-training external fit give 36 seed-level runs across the scratch D-MPNN and fine-tuned CheMeleon models. The colored vectors represent predictions for held-out molecules, not new training inputs.

a Many conformers, one equilibrium hydration target



b Summarize the supplied ensembles



c Predict six summaries from molecular structure



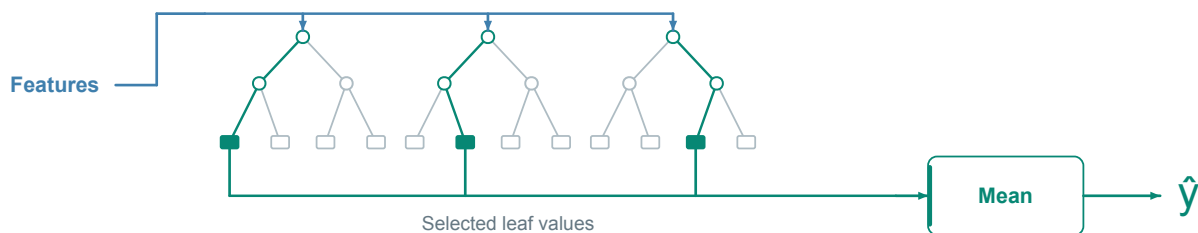
s_i : conformer solvation energy; i_0 : lowest-gas-energy conformer.

$\Delta g_i = g_i^{\text{water}} - g_i^{\text{gas}}$. Each phase uses its own energy minimum.

Supplementary Figure 7 | Conformer-resolved sources teach a molecular equilibrium response. **a**, Three illustrative conformers of 1,2-dimethoxyethane share one molecular equilibrium hydration target; they do not have separate experimental endpoint labels. **b**, Gas and water conformer energies are normalized separately to their phase-specific minima, giving g_i . The correction $C = -RT \ln \sum_i \exp(-g_i/RT)$ summarizes the supplied ensemble. Energy levels and their connecting lines are schematic, not measured ConfSolv energies. **c**, The six predicted summaries are the gas and water corrections, ensemble hydration minus the solvation energy of the lowest-gas-energy conformer, the population standard deviation of conformer solvation energies, and the arithmetic mean and population standard deviation of Δg_i . All energies are in kcal mol^{-1} . These summaries do not assert complete conformational sampling, and no conformer ensemble is regenerated at deployment.

a ExtraTrees

Partition feature space; average one stored leaf value from every tree.

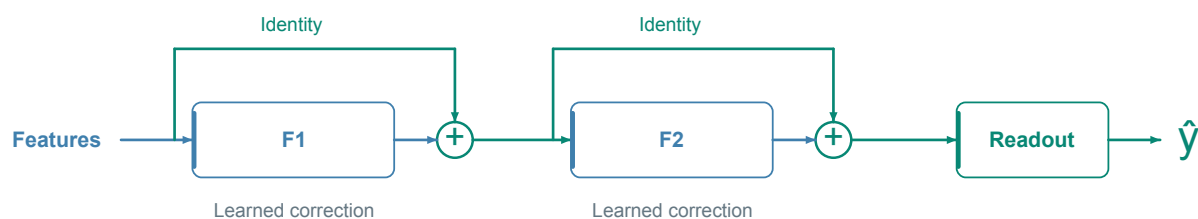


ARROW MAE 0.202 · Matched and shuffled-response controls

A finite range of leaf values bounds the prediction.

b Residual MLP

Learn nonlinear corrections along an identity-preserving pathway.

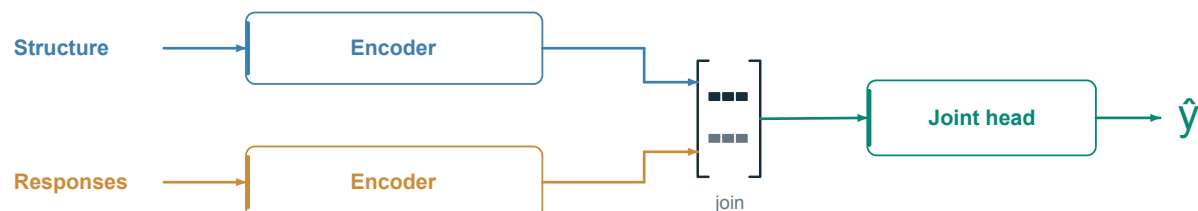


ARROW MAE 0.340 · Tests a direct neural replacement

A flexible function alone does not recover the tree accuracy.

c Dual structure / response branches

Learn a representation for each input group before combining them.



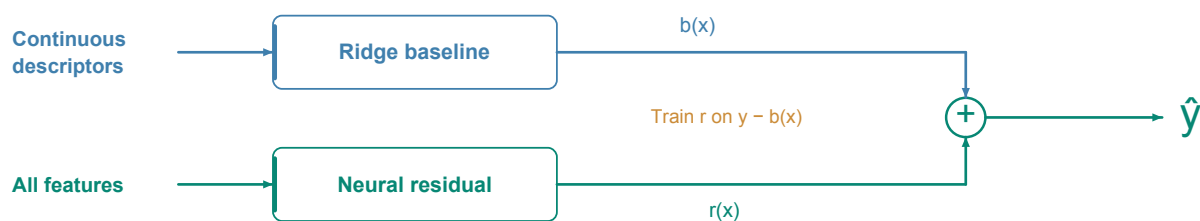
Size-held-out MAE 1.560 · Refitted ExtraTrees: 1.772

Separate branches do not impose physical additivity.

Supplementary Figure 8 | Local averaging and neural feature combination. **a**, ExtraTrees selects a leaf in each randomized tree and averages its stored value. **b**, A residual MLP learns corrections alongside identity paths. **c**, Separate structural and response branches are concatenated before a joint head. Illustrations explain mechanisms, not exact layer widths or tree counts. Numerical callouts refer to the evaluations in Table 2, not theoretical properties of the architectures. All mechanism diagrams use editable vector geometry; they are explanatory schematics, not data-derived plots.

a Ridge + neural residual

Combine a continuous baseline with a learned correction.



Size-held-out MAE 1.391 · No tree-leaf ceiling in the correction

A linear trend is not a validated asymptotic free-energy law.

b Size-sensitive message passing

Count and aggregate atom states, not only fragment presence.

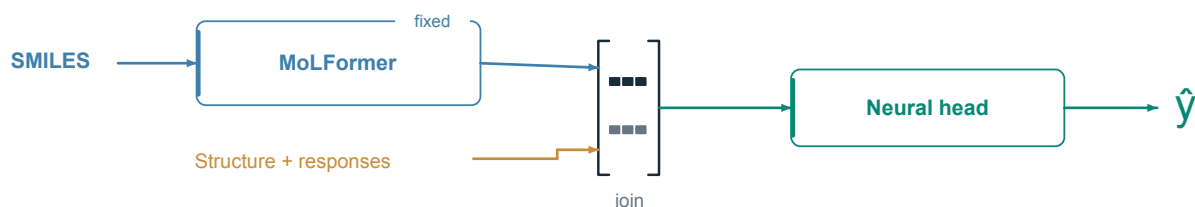


External MAE 1.168 · Without responses: 1.311

Shown: sum / mean / count readout; an atomic-plus-global variant was also tested.

c Frozen MoLFormer + neural head

Reuse a pretrained sequence representation alongside response information.



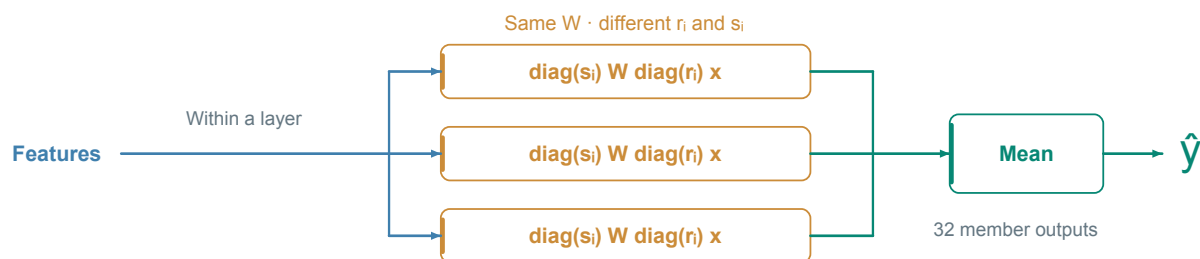
External MAE 1.111 · Responses help in all three cohorts

The encoder stays fixed; the endpoint learns from hydration labels.

Supplementary Figure 9 | Baselines, graph aggregation and pretrained encodings. **a**, Ridge predicts a continuous baseline and a network learns the remaining target, $y - b(x)$, during training; inference adds the baseline and predicted residual without access to the query's target. **b**, One graph variant combines summed and mean atom states with atom-count information; a separate atomic-plus-global variant is also tested. The summaries are concatenated, not numerically equated or treated as independent free-energy contributions. **c**, MoLFormer is frozen, while the downstream head learns to combine its representation with structural and response information. Schematics omit layer detail; all source surrogates remain fixed.

a TabM: a parameter-sharing ensemble

Use member-specific factors within layers that share their main weights.

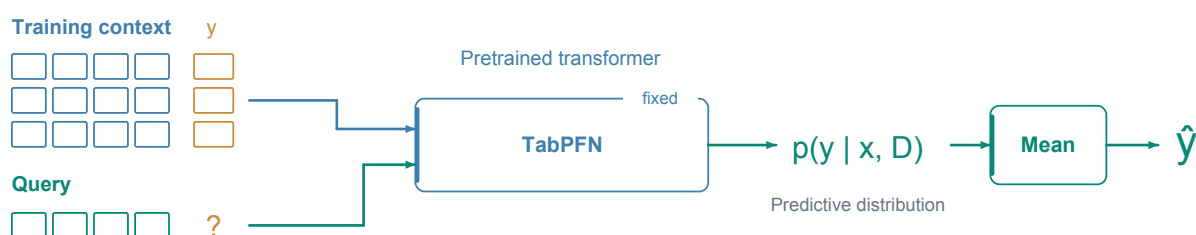


External MAE 1.132 · Size-held-out MAE 1.389

32 members share parameters; these are not 32 independent training runs.

b TabPFN-3.5: prediction from context

Condition a pretrained transformer on eligible labelled training rows.

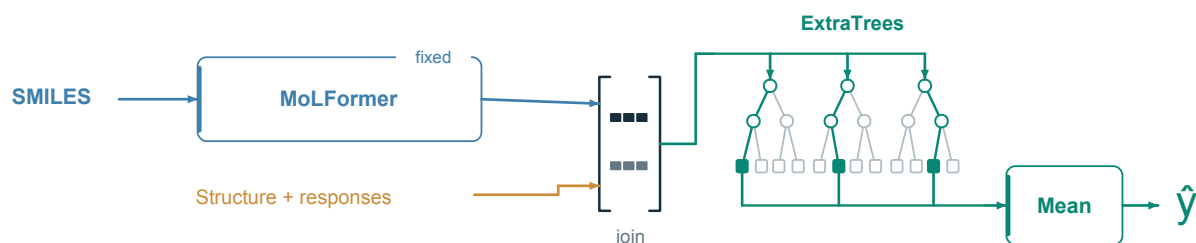


ARROW MAE 0.199 · Size-held-out MAE 1.318

Native intervals under-cover; input recipes differ across partitions.

c Frozen MoLFormer + ExtraTrees

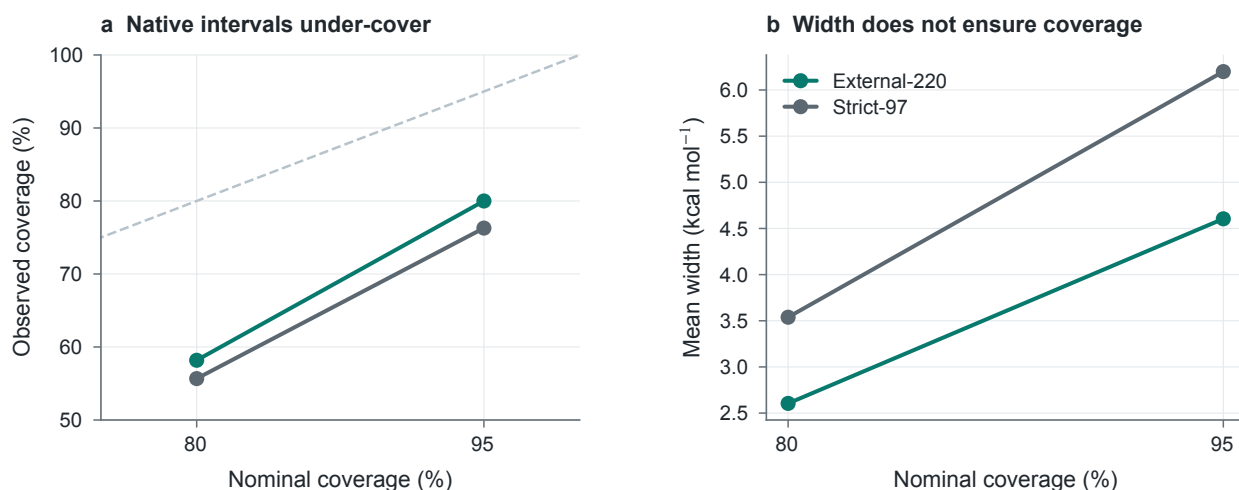
Change the representation while keeping the tree endpoint family.



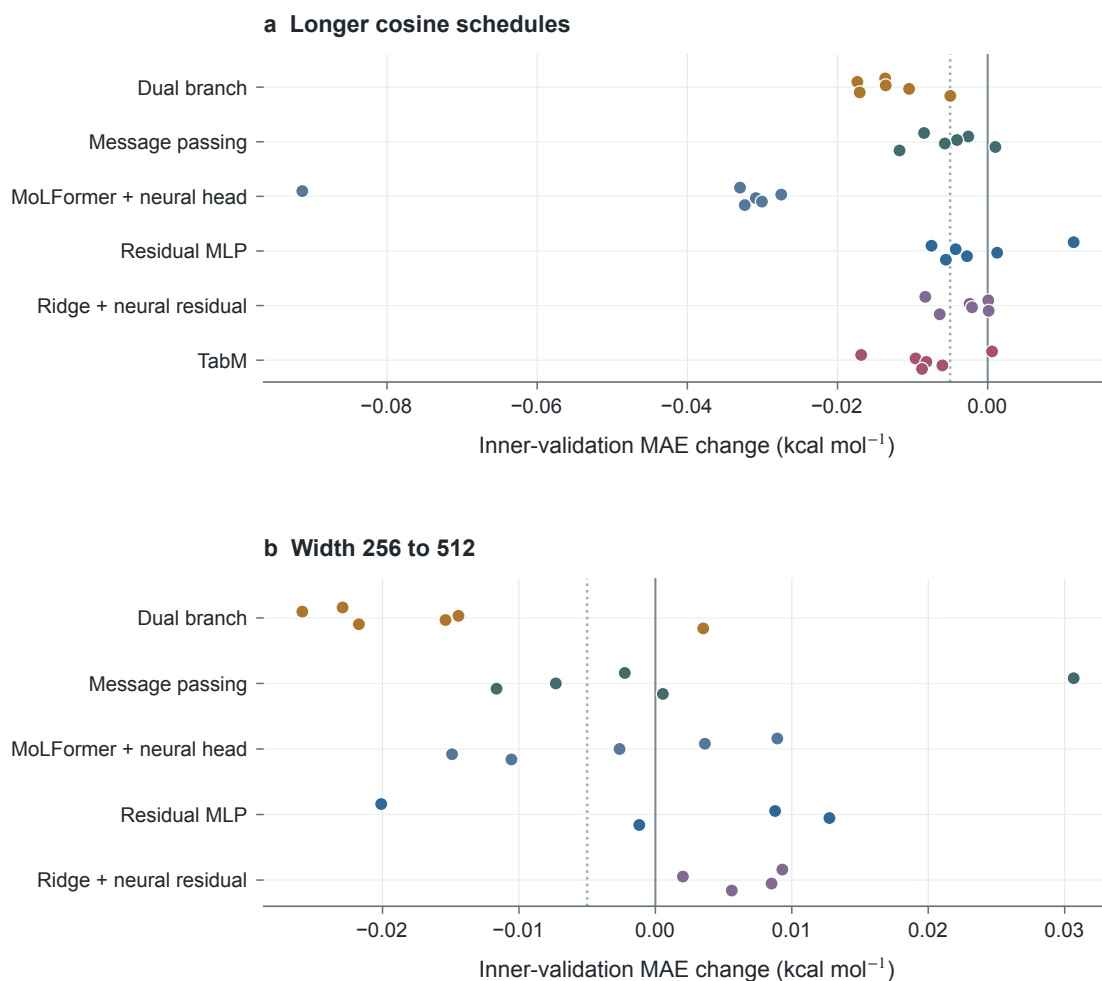
ARROW MAE 0.234 · Without responses: 0.348

The endpoint remains bounded by its stored leaf values.

Supplementary Figure 10 | Shared ensembles, in-context inference and a representation control. **a**, TabM shares weight matrices while member-specific input/output modulations within its layers produce 32 predictions. Three paths are shown for clarity; the mean is used. **b**, TabPFN conditions on labelled training rows and an unlabelled query using pretrained weights. No query label is supplied and no gradient fine-tuning is done here. Feature recipes vary by partition (S19). **c**, Frozen MoLFormer features are combined with structural and response features before an ExtraTrees endpoint, isolating a representation change while retaining leaf-value averaging. Three representative trees are shown, not the full ensemble.



Supplementary Figure 11 | Native predictive intervals require calibration. **a**, Empirical versus nominal coverage for the equal five-model TabPFN mixture; the dashed diagonal marks nominal coverage. **b**, Corresponding mean interval widths. Strict-97 is nested in External-220. These intervals are shown in teal for External-220 and gray for Strict-97 in both panels. They are evaluated without external calibration, and their undercoverage is not represented by seed dispersion. Connecting lines guide the eye between the two assessed levels.



Supplementary Figure 12 | Training duration and capacity challenges. **a**, Mean three-split inner-validation changes at seed 47 for the 1,200-epoch cosine challenge relative to the ordinary schedule, one dot per outer partition and family. **b**, Corresponding changes when width is expanded from 256 to 512 for qualifying configurations. Negative favors the challenge; the dotted line marks -0.005 kcal mol⁻¹. Retention also requires improvement in at least two of the three pairs. These are inner-validation scores, not test results, and the duration challenge also changes the learning-rate schedule.